# Can memistor-class electrochemical weights train neural networks?

A pre-registered simulation study from Widrow's 1960 memistor to modern ECRAM, with a device specification sheet and an experimental roadmap

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**Abstract**

Widrow and Hoff stored neural-network weights in memistors, three-terminal electrochemical cells, in 1960; lithium-ion, metal-oxide and protonic ECRAM are their modern descendants. We ask, in simulation, how close such devices come to digital training when every weight update is made of device pulses and the reads are kept ideal (fp32). We compare plain SGD and three Tiki-Taka-family algorithms on five device specifications, on MNIST, a CIFAR-10 subset and a byte-level language model, with five seeds that share initialisation and data order across arms. Against SGD with exact updates at the same learning rate, 4 pairings (Li-ECRAM and protonic-inspired, each with chopped TTv2 or AGAD) stay within the pre-set margins on all three tasks, while plain SGD exceeds them on every published device. Against a separately tuned digital Adam no pairing does: on the language model the best analog result is 23.36 % worse in bits per byte. Over a factor of four in width no arm's gap grows detectably beyond the margin; endpoint equivalence was established for one CIFAR pairing only, and an analysis paired by seed detects growth for another without resolving its size against the margin. A sensitivity sheet with three seeds establishes only a few one-at-a-time ranges and shows that an earlier asymmetry limit was mostly lost weight range. Against explicit pulse streams the aggregate update meets preset point-estimate lines on 10 of 12 probes (bootstrap: 1), not where clipping removes nearly all of an update, and ignores shared-line dependence; in training one label and two close rankings change. A conditional energy projection, borrowing an analog inference chip's read efficiency, gives 17.9–32.4× less energy per weight-row than an H100, for cells that pass the margin under ideal reads. This version repairs the simulator after independent reviews and reports how far each repair moved the original results.

## 1 Introduction

Training a neural network on a digital processor moves every weight from memory to arithmetic and back at every update. In 1960, Widrow and Hoff took a different route: they stored each weight of their adaptive ADALINE neuron in a *memistor*, an electrochemical cell whose conductance is set by the time integral of a current through a third terminal [1–3]. The weight was read without being disturbed, and it was adapted in place. The device was macroscopic, the cells were wet, and no multilayer learning rule existed yet, so the line ended with the Madaline hardware of the early 1960s [3]. That machine was a working analog learning system.

Sixty years later, three-terminal electrochemical devices have returned at the nanoscale. Lithium-ion ECRAM [4, 5], CMOS-compatible metal-oxide ECRAM [6], protonic programmable resistors [7, 8] and open-loop programmable arrays [9] all decouple the read path from the write

path, just as the memistor did. They report hundreds to thousands of conductance levels, nanosecond pulses and femtojoule-scale write energies. In parallel, algorithms for training on imperfect analog weights have matured: the stochastic pulse-coincidence update of the resistive processing unit (RPU) [10], Tiki-Taka [11] and its successors TTv2 [12], chopped TTv2 and AGAD [13], and residual learning with convergence guarantees [14].

That electrochemical weights can train small networks is already established: Chen et al. trained in situ on an ECRAM array and simulated larger networks [9], and Rasch et al. studied reference correction, retention and device requirements for the TT family [13]. Most recently, Wu et al. [15] simulated ECRAM-calibrated Transformers of up to about 123 million parameters with finite converters and device nonidealities, keeping the weight-gradient path digital. Our contribution is narrower and specific: *a reproducible, seed-matched simulation comparison* of plain SGD and three device-tolerant algorithms on a catalog of published (or, where nothing is published, labelled hypothetical) electrochemical devices, with the weight update done entirely in the devices, on three small tasks; a sensitivity analysis against absolute accuracy lines; and a *conditional* energy projection. Results are stated against two digital baselines: SGD with exact updates at the same learning rate, and a separately tuned Adam.

What this paper provides:

1. **A simulator with stated limits.** Reads are ideal (fp32); only the update is analog. It reproduces selected published outcomes of Gokmen and Haensch 2020 under this simplified read model, and the offset-robustness ordering of Rasch et al. 2024 (Section 4.1). After an independent review it was repaired (separate random streams, a reset made of pulses, the AIHWKIT device presets mapped one to one) and each repair is tested (Section 3, Appendix F).
2. **A seed-matched comparison** of SGD, TTv2, chopped TTv2 and AGAD on five device specifications, three tasks (MNIST, a CIFAR-10 subset, a byte-level language model on human-written text) and five seeds, with every arm of a seed sharing its initial weights and data order (Section 4.2).
3. **A width study** with per-arm learning rates and an equivalence test (Section 4.4).
4. **A sensitivity sheet** against absolute accuracy lines, with the asymmetry sweep repeated at a fixed symmetry point so that asymmetry is not confounded with lost weight range (Section 4.5).
5. **A conditional energy projection** for the weight-matrix operations, reported only where the device reaches the stated quality (Section 5).
6. **A proposed roadmap** of device experiments and a proposed, unbuilt bench replication protocol of the 1960 memistor (Section 7, Appendix A).

Every experiment was planned in a committed plan before it ran; the plans are internal, timestamped by the repository, not an external registry. The original runs and the revision runs are kept apart, and every deviation is recorded (Section 9). All results are simulations, and every device parameter carries a provenance label (measured, fitted, claimed, estimated or assumed).

## 2 Background and related work

### 2.1 The memistor (1960)

Widrow and Hoff introduced the ADALINE and the least-mean-squares (LMS) rule in 1960 [2]. To store its weights they built the memistor [1]. A pencil-lead (graphite) substrate sits in a copper-

sulphate plating bath. Copper is plated onto it or removed from it by a current through a third electrode, and the conductance between the lead's two ends is sensed with a small AC signal that does not disturb the state. Widrow, Pierce and Angell described elements with "thousands" of analog levels that spanned roughly 100 to 1 ohm in about ten seconds at a few milliamperes; this description is reported in secondary sources, see [16]. Mouttet analyses the memistor as a three-terminal memristive system and summarises Widrow's ideal: a conductance linear in the plating charge, reversible, non-volatile and free of hysteresis [16]. Widrow and Lehr recount that the Madaline I machine was built in hardware with memistor weights and used for pattern recognition research [3]. Write noise, up/down asymmetry, retention and endurance were never reported in a form we could find, so in our catalog these are estimates (Section 3.4). The memistor predates Chua's memristor [17] and its later physical identification in titanium-oxide films [18]. It is the conceptual ancestor of today's three-terminal synaptic transistors.

### 2.2 Device requirements for training

Gokmen and Vlasov [10] defined the resistive processing unit (RPU). Updates are stochastic pulse trains whose coincidences change the conductance by a fixed step (stochastic computing, [19]). They derived device specifications for plain SGD on MNIST at a small error penalty: about a thousand levels, up/down asymmetry of a few percent, and tolerance to large cycle-to-cycle and device-to-device variation. Agarwal et al. [20] reached similar conclusions; the requirements were extended to convolutional networks [21], and reviews cover the device landscape [22–25]. Mixed-precision schemes [26] accumulate the gradient digitally and program the devices rarely. They relax the device demands but give up the fully parallel in-memory update.

### 2.3 Algorithms that tolerate asymmetric devices

Real devices change their conductance by different amounts up and down, and the step depends on the state (soft bounds). Under plain SGD this acts as an implicit penalty that pulls weights toward each device's symmetry point [11, 14].

- **Tiki-Taka (TTv1)** [11] uses two arrays. The gradient is accumulated on an array $A$, and its columns are periodically transferred into the weight array $C$, with $W = \gamma A + C$. Asymmetry then acts as a useful decay on $A$ instead of a bias on $W$.
- **TTv2** [12] inserts a digital buffer $H$ between the two arrays. $H$ low-pass filters the reads of $A$, and a pulse is sent to $C$ only when a threshold is crossed. TTv2 still needs every device's symmetry point to be subtracted by a reference.
- **Chopped TTv2 (c-TTv2)** and **AGAD** [13, 27] remove this sensitivity. A random sign modulates the inputs and the reads (c-TTv2). AGAD additionally replaces the static reference with a digital, on-the-fly reference taken from the recent history of $A$.
- **Residual learning** [14] gives convergence guarantees for this family.

IBM's open-source AIHWKIT [28] implements these algorithms with fitted device presets, including the Li-ECRAM and metal-oxide ECRAM presets whose parameters we use. In the revision these two presets are mapped one to one onto AIHWKIT's own LinearStepDevice model, with its shifted bounds and slope spread, and checked against AIHWKIT's pulse responses (Section 3.4).

### 2.4 Electrochemical three-terminal devices

- **Lithium-ion synaptic transistors and ECRAM.** Fuller et al. [5] showed a lithium-ion synaptic transistor, followed by an ionic floating-gate array [29]. Tang et al. [4] reported Li-$WO_3$ ECRAM with about a thousand symmetric, linear states and nanosecond pulses.
- **Metal-oxide ECRAM.** Kim et al. [6] made it CMOS-compatible.
- **Protonic resistors.** Onen et al. [7] reported nanosecond protonic programmable resistors with a phosphosilicate-glass electrolyte. In 2026, Xu et al. [8] quantified cycle-to-cycle (3%) and device-to-device (21%) variability of BEOL-compatible protonic ECRAM.
- **Open-loop programmable arrays.** Chen et al. [9] demonstrated a $10 \times 10$ W/YSZ/$WO_x$ array with 256 states.
- **Organic devices.** Organic electrochemical synapses [30] extend the family to low voltages.

In-situ training on ECRAM hardware has so far been shown on small arrays, for example a $3 \times 3$ array by Li et al. [31] and the $10 \times 10$ array of [9]. Other memory technologies have gone further. Phase-change and resistive arrays have trained or run networks at larger scale [32–35], and analog-AI inference chips with on-chip periphery now report measured efficiencies of about 10 TOPS/W [36, 37].

**Scaling analog training.** Wu et al. [15] co-designed the weight mapping, the optimizer and the converters, and simulated ECRAM-calibrated Transformers up to about 123 million parameters with finite converter resolution and device nonidealities. Their weight-gradient path stays digital, so they do not test the fully in-memory update studied here. The two studies answer complementary questions: what an analog update saves in computation and data movement, and what it costs in accuracy and calibration. Our models are far smaller (Section 6).

## 3 Methods

### 3.1 Device model

Each analog weight is one device with a normalised state $u \in [-1, 1]$ between its own bounds. Its conductance is $g = g_{\min} + \frac{u+1}{2}(g_{\max} - g_{\min})$, and the weight is read differentially as $w = \alpha\,(g - g_{\text{ref}})$. One nominal pulse moves $u$ by $k = 2/n_{\text{states}}$ at mid-range. With asymmetry $a$ the up and down steps are $k_\pm = k\,s\,(1 \pm a/2)$, where $s$ is a per-device step factor with spread $\sigma_{\text{d2d}}$. With saturation (soft bounds) the steps depend on the state:

$$\Delta u_+ = k_+\,(1 - \gamma_+ u), \qquad \Delta u_- = -k_-\,(1 + \gamma_- u). \tag{1}$$

The per-device slopes $\gamma_\pm$ have a relative spread $\sigma_\gamma$. A device's symmetry point, where an up and a down pulse cancel, is $u^* = (k_+ - k_-)/(k_+\gamma_+ + k_-\gamma_-)$. The reference $g_{\text{ref}}$ is set at $g(u^*)$ plus a calibration error of standard deviation $\sigma_r$, in units of the half-range. A constant-step device has no symmetry point and is referenced at mid-range.

- **Several pulses in one update.** When a device receives $n_+$ up and $n_-$ down pulses in one update, the pulses are taken as uniformly interleaved, and $\mathrm{d}u/\mathrm{d}s = A - Bu$ is solved exactly over the batch, with $A = k_+n_+ - k_-n_-$ and $B = k_+\gamma_+n_+ + k_-\gamma_-n_-$.

- **Write noise.** Cycle-to-cycle noise adds a Gaussian with standard deviation $c_{2c}$ times the local step, growing as the square root of the pulse count.
- **Drift.** Retention loss relaxes $u$ toward the bare state $u = -1$ with a time constant $\tau$ in training steps.
- **Bounds.** The bounds $g_{\min}$, $g_{\max}$ carry a device-to-device spread.

**The AIHWKIT LinearStepDevice (revision).** The two devices taken from AIHWKIT presets (Li-ECRAM, MO-ECRAM) use AIHWKIT's own parameterisation instead: the state is the weight itself, with hard bounds $b_{\max} = w_{\max}(1 + \sigma_b\xi)$ and $b_{\min} = w_{\min}(1 + \sigma_b\xi)$ around shifted means ($w_{\max} = 1.1724$, $w_{\min} = -0.8276$ for Li-ECRAM, so that the symmetry point is at zero), steps $k(1 \pm \beta + \sigma_{\mathrm{d2d}}\xi)$, and slopes divided by the mean bounds, $\Delta w_+ = k_+(1 - |\gamma_+ + \sigma_{\gamma+}\xi|\, w/w_{\max})$ and $\Delta w_- = -k_-(1 + |\gamma_- + \sigma_{\gamma-}\xi|\, w/|w_{\min}|)$, with multiplicative write noise. The original version of this paper copied the preset fields into the model above with bounds $[-1, 1]$ and no slope spread; that mapping is kept only for comparison (Appendix F).

Forward and backward passes read the weights ideally, in fp32. Read noise and DAC/ADC quantisation are outside the scope of this study, which isolates the *update*; the energy projection (Section 5) nevertheless prices finite-precision reads, so accuracy and energy rest on different read assumptions. Biases, layer norms and the language model's position table are digital.

### 3.2 The update

We use the stochastic pulse-coincidence scheme of [10]. Each sample (row) offers $BL = 31$ pulse slots. Input $i$ fires with probability $p_{x,i} \propto |x_i|$ and error $j$ with probability $p_{d,j} \propto |d_j|$, with the scaling split equally between the two streams (update management). Weight $ij$ therefore receives coincidences with mean $\eta\,|x_i d_j|/(\alpha k)$. Up and down counts are kept separate, so asymmetry is never cancelled digitally. Within a batch the counts of all samples are summed and drawn from a Poisson distribution with the requested mean. A mini-batch therefore only groups pulses; it does not average gradients.

This is a surrogate, not a pulse-by-pulse simulation. The mean is exact only while the largest pre-clamp slot probability is at most one, i.e. $(\eta/\alpha k)\max|x|\max|d| \le BL$; beyond that the probabilities are clipped and part of the update is lost. The repaired simulator records, per layer, the fraction of updates that clipped and the lost fraction of the requested update (Section 4.2). Discrete pulses do not commute on a state-dependent device (on a 20-level soft-bounds device one up then one down pulse from $u = 0.4$ ends at 0.314, the reverse order at 0.334, and the batch solution at 0.327), and independent per-cell Poisson draws ignore the correlations of shared row and column pulse streams. A larger variance than the binomial one is therefore not claimed to be conservative. Section 4.3 compares the surrogate with explicit pulse streams on training cases.

### 3.3 Algorithms

**SGD.** The pulses go directly into the weight devices.

**TTv1 [11].** The pulses go into an array $A$, and the forward weight is $C + \gamma A$. Every batch, $n_c$ columns $j$ of $A$ are read and added to $C$ by a pulsed outer product, $\lambda A_{:,j}\, e_j^\top$. $A$ is never reset.

**TTv2 (local variant of [12]).** Each read column is added to a digital buffer, $H_{:,j}$ += $\lambda A_{:,j}/(\alpha k)$. $C$ receives up to 31 pulses where $|H| \ge 1$, and $H$ keeps the remainder. Optionally the read column of $A$ is reset after the read. Published TTv2 uses single-pulse transfers, forgets a buffer

element after it fires and does not reset $A$; we name ours a variant and run the published-style form beside it in the diagnostic (Appendix F).

**The reset of $A$ (revision).** The original simulator moved the read column of $A$ straight to its symmetry point and charged $2BL = 62$ pulses it never applied (the *ideal reset*). Applied through the device model, those 62 alternating pulses leave 97.55 % of the displacement on the protonic-inspired device. The repaired *pulse reset* uses the transfer's own read: a sign-directed train of round($|A_{ij}|/\alpha k$) pulses toward the read zero, re-read and corrected once, at most 62 pulses per weight, every pulse applied through the device dynamics (with its write noise, state-dependent step and device spread) and counted. The ideal reset is reported only as a labelled variant.

**c-TTv2 [13].** Chopper signs $c_j \in \{\pm 1\}$ modulate the inputs of the $A$ update ($c \odot x$) and demodulate the reads ($c_j A_{:,j}$). Each read column's chopper flips with probability $\rho$.

**AGAD [13].** The choppers of c-TTv2, plus a digital leaky average of the reads, $p_j \leftarrow (1-\beta)p_j + \beta A_{:,j}$. The read is $z_j = c_j(A_{:,j} - p_j^{\text{ref}})$, and $p_j^{\text{ref}}$ is reset to $p_j$ whenever $c_j$ flips.

In the main experiments every TT-family $A$ array uses a learning rate $1/\lambda$ times larger, so that $\lambda\times$ (the $A$ rate) equals the SGD rate; the TTv1 configuration of the published benchmark (Section 4.1) is the exception ($\eta = 0.01$, $\lambda = 0.02$, $A$ rate equal to $\eta$). The settings, including the choice between the pulse reset and no reset, were chosen once on MNIST validation with a mid-quality device that is not in the catalog (Section 3.6; Appendix B). We implemented the algorithms from the papers' text; we did not use the authors' code.

### 3.4 Device catalog and provenance

Table 1 lists the devices as every revised run uses them. Every field carries a provenance label:

- MEASURED: reported in the paper;
- FIT: from the AIHWKIT presets fitted to the published devices [28];
- CLAIM: a qualitative statement in the paper, translated by us;
- ESTIMATE: ours;
- ASSUMED: for example, the 2% reference error given to every device.

The protonic device of [7] is mostly estimated: the paper reports a thousand up and a thousand down pulses that are "nearly linear and symmetric", but no write-noise or variability statistics, and its finite retention is replaced by no drift. We therefore call it a *protonic-inspired scenario*; a thousand applied pulses are not a demonstration of a thousand independently programmable states. The original field copy of the Li-ECRAM and MO-ECRAM rows is kept, labelled as superseded, in Table 17 (Appendix E); the revised runs use the one-to-one AIHWKIT mapping above (bounds $[-0.8276, 1.1724]$ and $[-0.8286, 1.1714]$, absolute slope spread 0.05). Widrow's memistor is given two specifications, one as he described it and one pessimistic, together with a pencil-lead cell built today with modern hobby materials (Appendix A).

Table 1: Revised device catalog, the configuration of every revised run (simulator units; provenance per field in records/revision/CATALOG-REV.json). * AIHWKIT's presets mapped one to one onto its LinearStepDevice (shifted bounds, bound-scaled slopes, device spread 0.05 of the slopes and 0.02 of the asymmetry); the others use the paper's model with bounds $[-1, 1]$. Columns: levels $n_{\text{st}}$, asymmetry $a$ (device spread), saturation $\gamma_+/\gamma_-$, bounds, write noise $c_{2c}$, step spread $\sigma_{\text{d2d}}$, reference error $\sigma_r$; no drift. The original field-copy catalog is Table 17 (Appendix E).

| Device | Label | $n_{\text{st}}$ | $a$ | $\gamma_+/\gamma_-$ | bounds | $c_{2c}$ | $\sigma_{\text{d2d}}$ | $\sigma_r$ |
|---|---|---|---|---|---|---|---|---|
| Li-ECRAM* | PUBLISHED-FIT | 1000 | 0 | 0.12/0.51 | [-0.83, 1.17] | 0.3 | 0.1 | 0.02 |
| MO-ECRAM* | PUBLISHED-FIT | 7089 | 0 | 0.42/0.73 | [-0.83, 1.17] | 2 | 0.1 | 0.02 |
| Protonic-inspired | PARTLY-PUBLISHED | 1000 | 0.05 | 0.20/0.20 | [-1, 1] | 0.3 | 0.1 | 0.02 |
| W/$WO_x$ ECRAM | PUBLISHED-FIT | 256 | 0 | 0.63/0.63 | [-1, 1] | 2.048 | 0.023 | 0.02 |
| Memistor (optimistic) | CLAIM+ESTIMATE | 1000 | 0.05 | 0.10/0.10 | [-1, 1] | 0.3 | 0.2 | 0.02 |

### 3.5 Tasks and baselines

**MNIST [38].** A 784–256–128–10 tanh MLP; 10 epochs, batch 128, on 55,000 training images. 5,000 held-out training images form the validation set, and the 10,000 test images are the test set.

**CIFAR-10 subset [39].** Three $3 \times 3$ convolutions with $c$, $2c$ and $2c$ channels (base $c = 32$), computed as crossbar matrix products over unfolded patches, then two fully connected layers. 30 epochs, batch 128, on the first 10,000 images of training batches 1–2. The next 2,000 images are the validation set, and the 10,000-image test set is the test set.

**Character LM.** A byte-level, two-layer transformer [40] of width $d$ (base 128) and context 128, trained on 8 MB of enwik8 [41], which is human-written Wikipedia text of 2006. 3,000 steps of $32 \times 128$ bytes. Every learned weight matrix uses the device model, including the embedding and the head; the activation-dependent attention products and the other operations are digital. Validation is the slice's last 0.5 MB, held out from training in selection runs. Test is a separate 0.5 MB slice.

The first baseline ("FP-SGD") is the same network, initialisation, data order and update rule with an exact floating-point weight update. The second is Adam [42] on digital twins with the same initialisation and data order and its own learning rate selected on validation; in the revision it is reported for all three tasks, beside FP-SGD. The gap is FP accuracy minus device accuracy for classification, in percentage points. For the LM it is the device's bits per byte relative to FP's, minus one, in percent.

### 3.6 Protocol and statistics

Every experiment was pre-registered in a committed plan before it ran (Section 9).

- **Hyperparameters** were chosen on validation splits only, with an edge rule: whenever the best value of a grid lies at its edge, the next half-decade point is added. The learning rate of every device arm in the main table is the FP validation choice. The scale study instead tunes each arm separately.
- **Statistics.** The main table uses five seeds per arm. Gaps are paired by seed, and we report the mean and a 95% Student-$t$ interval. A cell PASSES when the whole interval lies below

the gap line (2 pp for classification, 5% for the LM), FAILS when it lies wholly above, and is UNRESOLVED otherwise.

- **Multiplicity.** We make no correction for multiplicity. The intervals describe each cell, not the family, and a best cell chosen among many is optimistic.
- **Random streams (revision).** Each seed defines separate generators for the logical initialisation, the data order, each tile's device population and the pulse and noise draws, so the paired gap compares runs that share their initial weights and their data.
- **Width (revision).** The endpoint contrast $D$ (gap at the widest minus the narrowest width) is tested for equivalence (TOST at 5 %, margins equal to the gap lines); "no evidence of growth beyond the margin" when only the upper bound is under the margin; never "flat".
- **Sensitivity (revision).** Each grid point is labelled against the *absolute* line with 3 seeds paired with FP-SGD of the same seed; a threshold is the weakest value that passes with every value toward the best end passing.

### 3.7 Simulator checks

The simulator is checked by 14 checks, and each check is shown to fail on at least one deliberately planted fault (17 faults in all). Examples of what the checks cover:

- the exact-update arm equals PyTorch SGD;
- the mean pulse update equals the SGD step;
- alternating pulses converge to the analytic symmetry point, including with per-device slopes;
- write noise grows as $\sqrt{n}$;
- drift relaxes to the bare state;
- a static read offset is integrated by TTv2, mostly cancelled by c-TTv2 and nearly fully cancelled by AGAD;
- TTv1 moves $C$ along $+\lambda A$;
- the SGD and TTv2 paths are bit-identical to the earlier code that produced our preliminary results.

The planted faults include a swapped pulse sign, a dropped chopper demodulation and a never-updated AGAD reference. Both suites run at the start of every compute session and stop it on any failure (Appendix C). The revision adds 8 checks of the repairs (matched initialisation and data order across arms, a shared device population across algorithms, resets that charge exactly the pulses they apply, the AIHWKIT LinearStepDevice recurrence, the clipping statistics, the published-style buffer, the re-centred symmetry point, and a bit-exact replay of the original simulator), each shown to fail on the old behaviour or a planted fault (Appendix F). These checks test the implementation; they do not validate the physical model.

# 4 Results

## 4.1 Simulator validation

**Gokmen and Haensch 2020.** We map their RPU-baseline device onto our model:

- step $10^{-3}$ at $w = 0$;
- mean saturation slope 1.66 with a 25% per-device spread;
- 30% step spread and 30% cycle-to-cycle noise;
- a symmetry point at $w = 0$ for every device.

We train their 784–256–128–10 sigmoid network on MNIST for 50 epochs at their learning rate of 0.01. The results are in Table 2 and Fig. 1a.

- FP reaches 1.78% test error, against their 2.0%.
- SGD on the device reaches 14.47% [12.68, 16.26], against "about 15%".
- Tiki-Taka reaches 2.26% [2.13, 2.39], against "close to 2%".

By our pre-registered criteria the agreement is **quantitative**: this is agreement with selected published outcomes under a simplified read model, not a full quantitative validation. Matching an endpoint under different sources of error cannot rule out compensating differences, and the batch-64 result below shows how sensitive the SGD arm is to the protocol. These validation runs used the original simulator (TTv1 has no reset, so the reset repair does not touch them). The known differences are listed in Section 6.1:

- we use mini-batches of 16, not 1;
- we do not model read noise, ADC/DAC quantisation or analog biases.

At batch 64 the FP and TTv1 results barely move, but SGD on the device degrades further. Summing more samples' pulses before the weight changes hurts a strongly asymmetric device, which is another reason to use batch 16 as the closer match to their batch 1.

**Rasch et al. 2024, qualitatively.** We use a 20-state soft-bounds device with an uncorrected reference offset of $\sigma_r = 0.5$ (Table 3, Fig. 1b).

- TTv2's test error rises from 4.12% to 24.27%.
- c-TTv2 rises less, from 4.29% to 13.02%.
- AGAD reads 8.47% versus 7.88% (no change detected with two seeds).

This is the published ordering; our pre-registered check reads **reproduced**. It rests on two seeds per condition and an ordering, so it does not establish that our c-TTv2 and AGAD are equivalent to the published implementations, and the absence of a detected change does not show that AGAD is unaffected. In our implementation AGAD starts from a higher error at zero offset on this very coarse device.

Table 2: Validation against Gokmen and Haensch 2020 (FCN 784-256-128-10, MNIST, 50 epochs): test error in %, our mean and 95% interval over 3 seeds at batch 16, and one seed at batch 64.

| Arm | Published [11] | Ours (batch 16) | Ours (batch 64) |
|---|---|---|---|
| FP (exact update) | 2.0 | 1.78 [1.75, 1.81] | 1.79 (n=1) |
| SGD on the RPU-baseline device | about 15 | 14.47 [12.68, 16.26] | 24.32 (n=1) |
| Tiki-Taka (TTv1) | close to 2 | 2.26 [2.13, 2.39] | 2.07 (n=1) |
| TTv2 (ours, 16 columns per batch) | – | 2.59 [2.42, 2.75] | not run |

Table 3: Qualitative check of Rasch et al. 2024 (Fig. 5): a 20-state soft-bounds device with an uncorrected reference offset $\sigma_r$; test error in % (mean and 95 % $t$ interval, $n = 2$; 10 epochs). FP: 2.69 [0.47, 4.92].

| Offset | TTv2 | c-TTv2 | AGAD |
|---|---|---|---|
| $\sigma_r = 0$ | 4.12 [2.79, 5.46] | 4.29 [3.91, 4.67] | 8.47 [7.58, 9.36] |
| $\sigma_r = 0.5$ | 24.27 [18.23, 30.30] | 13.02 [10.99, 15.05] | 7.88 [3.49, 12.26] |

**The device presets against AIHWKIT itself (revision).** On a rented machine (never on our desk) we installed AIHWKIT 1.1.0 from PyPI and drove its EcRamPresetDevice and EcRamMOPresetDevice through long up, down and alternating pulse schedules (500 devices each, with and without cycle-to-cycle noise), using its own plotting utilities. Our AIHWKIT-mode tile, driven through the same schedules, matches AIHWKIT's mean responses, device spreads and reached bounds within the pre-registered tolerance for both presets (largest mean difference 0.0111 for Li-ECRAM and 0.0113 for MO-ECRAM, in weight units; the pre-registered tolerance is $\max(4\,\mathrm{SE}, 2\,\Delta w_{\min})$ at every stored pulse). The original field-copy mapping differs by up to 0.174 for Li-ECRAM, mostly at the bounds (Appendix F).

### 4.2 Main comparison on the repaired simulator

Tables 4 and 5 give the gap of every device–algorithm pairing over five seeds, against FP-SGD (the same network, learning rate, initialisation and data order with an exact update) and against digital Adam at its own selected learning rate. Every arm of a seed now shares its initial weights and its data order (checked by hash in every record). The FP-SGD learning rate is reused for every device arm: this is a restricted question (what happens when the FP-SGD rate is reused), not a fair-tuning comparison of the algorithms; Section 4.4 tunes each arm separately.

**Against FP-SGD.**

- **Plain SGD exceeded the line** in 12 of the 12 cells of the four published devices, at the FP-selected learning rate.
- **All three tasks:** 4 pairings pass on every task: Li-ECRAM c-TTv2, Li-ECRAM AGAD, Protonic-inspired c-TTv2, Protonic-inspired AGAD. No TTv2 pairing does: with the pulse reset, TTv2 fails on CIFAR on every published device.
- **Per task,** 14, 6 and 4 of the 20 device cells pass on MNIST, CIFAR and the LM. W/YSZ/$WO_x$ fails on CIFAR and the LM with every algorithm, and MO-ECRAM fails on the LM.

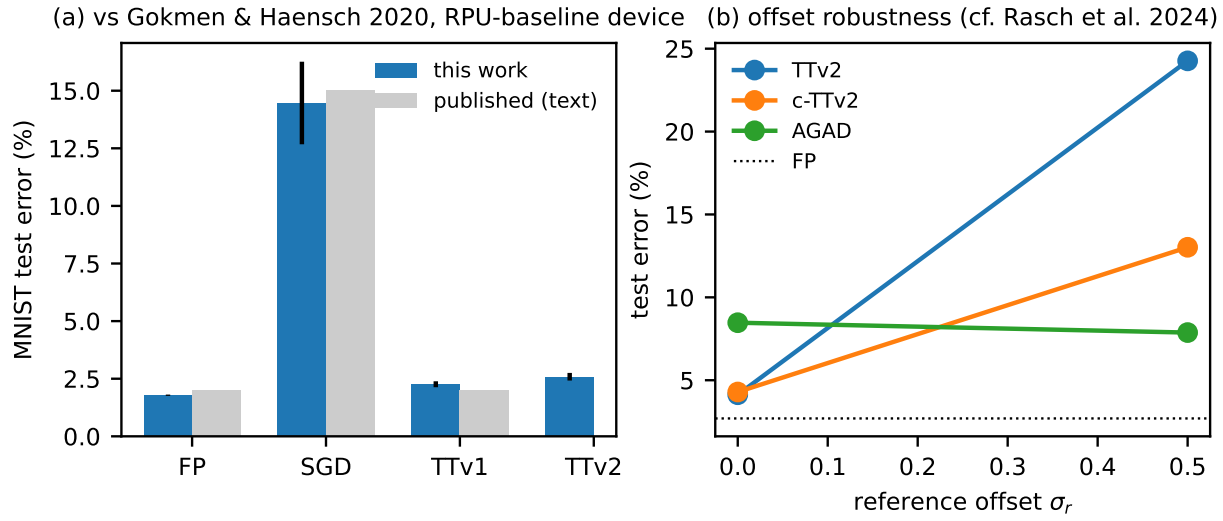



Figure 1: Simulator validation. (a) Test error of our simulator (mean and 95% interval, 3 seeds) against the values stated in the text of Gokmen and Haensch 2020 for their FCN on MNIST with the RPU-baseline device. (b) Robustness to an uncorrected reference offset on a 20-state device (2 seeds), the qualitative trend of Rasch et al. 2024, Fig. 5.

- **Best cells:** Memistor (optimistic) c-TTv2 on MNIST (0.18 pp), Protonic-inspired AGAD on CIFAR (-1.34 pp) and Protonic-inspired c-TTv2 on the LM (-1.53 %). A negative gap whose interval includes zero is not evidence of superiority; and because these are the best of many cells, they are optimistic.

**Against Adam.** Digital Adam is the stronger baseline. On the LM FP-SGD is 25.35 % worse in bits per byte than Adam, and **no analog cell comes within the 5 % line of Adam**: the best, Protonic-inspired c-TTv2, is 23.36 % worse (0 of 20 cells pass). On MNIST and CIFAR, where Adam and FP-SGD are close, 14 and 4 cells pass against Adam. No pairing passes all three tasks against Adam (0). The analog results are therefore close to the FP-SGD comparator on these tasks, not to the best digital training in this study.

**The margins.** The lines are practical non-inferiority margins, not equivalence: on MNIST a 2 pp loss from 97.77 % roughly doubles the error rate. UNRESOLVED means insufficient evidence against the margin, never adequate performance. The mean digital baselines are 97.77 % and 97.75 % on MNIST, 59.69 % and 60.76 % on CIFAR, and 2.965 and 2.366 bits per byte on the LM (FP-SGD and Adam). The FP-SGD seed standard deviation is 1.02 pp on CIFAR and 0.078 bits per byte on the LM.

**What the update loses.** Two properties of the repaired simulator matter for reading these results.

- **Clipping.** The TT family drives its $A$ arrays at ten times the SGD rate, which exceeds the $BL = 31$ pulse budget in the largest layers: the mean lost fraction of the requested $A$ update reaches 57.5 % in the CIFAR output layer and 27.8 % in the LM embedding (plain SGD at most 0.7 % in any layer's mean). These are means over the device runs of one layer;

the worst single layer record loses 96.4 % (MO-ECRAM AGAD, CIFAR output layer). The logged fraction is the discarded unsigned expected pulse mass, not the norm of the lost signed gradient after cancellation. Clipping the line probabilities at one is the update-management policy chosen here (pulse length and update management are configurable choices [28]), not a property of the devices; it caps the effective $A$ rate in those layers, and the aggregate update itself is an approximation (Section 4.3).

- **Reset residual.** With at most 62 pulses per weight, the pulse reset leaves a residual on the read $A$ columns. Averaged with every layer and run weighted equally (a macro-average of the per-layer means), it is 67.00 nominal steps; pooled over all reset events it is 26.97 (Li-ECRAM 3.7, protonic-inspired 5.5, MO-ECRAM 113.4); the largest layer mean is 2081.42 (MO-ECRAM). Reset pulses are 3.1 % of all pulses as the mean of the per-run fractions, and 0.49 % pooled (all reset pulses over all pulses). The macro-averages say what a typical layer and run see; the pooled numbers what the arrays receive in total.

**Ideal-device reference.** The same algorithms on an ideal device (symmetric, constant step, 1000 levels, no noise, no spread, exact reference) are the rows "Ideal 1000-level" of the tables (deviation R2). On MNIST every algorithm passes on it. On CIFAR and the LM none does: plain SGD loses 6.32 pp and 12.14 % (stochastic pulse updates on a 1000-level grid at the FP-SGD rate), and c-TTv2 and AGAD lose 18.04 and 19.20 pp on CIFAR, far more than on the saturating Li-ECRAM and protonic-inspired devices. With the settings selected here, the TT family relies on the device's own saturation as the decay of its accumulator; the ideal device has none. This is a property of these algorithm settings, not a general requirement that devices saturate: a sufficiently precise symmetric device can implement SGD, as the MNIST row shows. This reference keeps stochastic pulse counts, bounds, transfer thresholds and the selected update management, so it is not an exact-update, algorithm-only TT reference, and its MNIST success does not show that every other failure is an unavoidable device limitation.

### 4.3 Fidelity of the aggregate update

The simulator draws each batch's pulses as independent Poisson counts per cell and applies them in one exact step of the interleaved dynamics (Section 3.1). A physical stochastic-pulse array instead fires row and column lines per slot and per sample; a cell is pulsed when both of its lines fire, and the samples follow one another. After the second review we compared the two, pre-registered with pass lines (`PLAN-REVISION-2.md`, experiment A), on the operating regime of the main table.

**Update level (A1).** At two points of training, one batch's inputs, errors and array state were taken from surrogate training runs and applied repeatedly from the same state by three methods: the surrogate; the explicit stream applied pulse by pulse (Bernoulli line firings, the per-pulse device step, write noise and bounds after every pulse; the literal stream); and a per-sample form of the same stream (seq), which replays the literal stream's line firings. Within one sample every pulse of a cell has the same sign, so seq applies the sample's train in closed form. Its write noise is an approximation: it is added once per sample at the pre-sample state (standard deviation $c_{2c}\sqrt{n}$ times the local step for $n$ pulses), and the bounds are applied after each sample, where the literal stream applies noise and bounds after every pulse. The cases cover clipped and unclipped layers: MNIST Li-ECRAM c-TTv2 (first and output layer), CIFAR MO-ECRAM AGAD (fully connected and output layer; the worst-clipped layer of the main table) and the LM protonic-inspired c-TTv2 (embedding and output layer), 12 probes. Version 3 compared seq with the literal stream as if

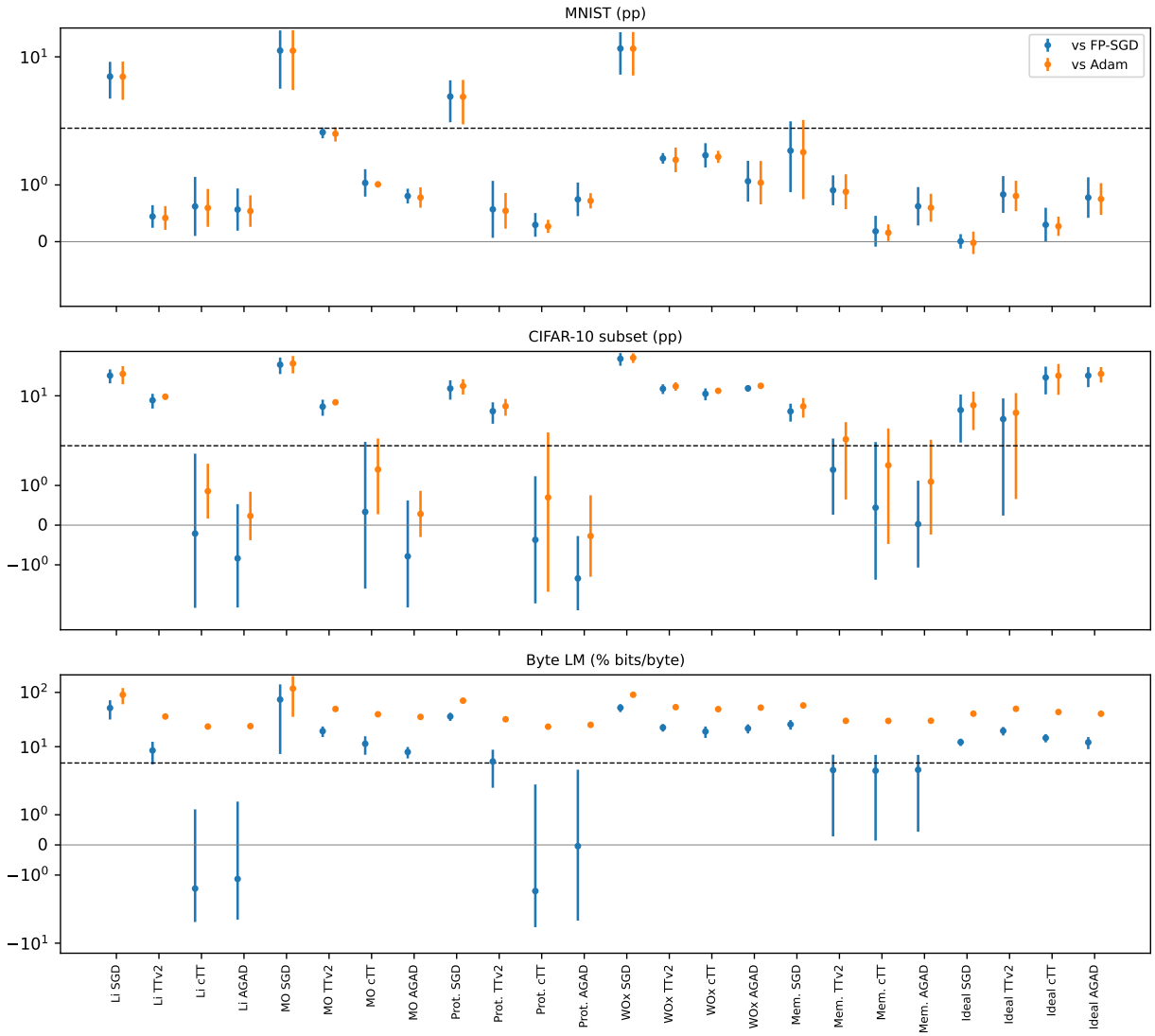



Figure 2: Repaired simulator: gap to FP-SGD (blue) and to digital Adam (orange) per device and algorithm, mean and 95 % interval over 5 seed-matched runs (symmetric-log axis); the dashed line is the gap line.

their draws were independent, although they share the line firings, so its sampling floor could hide a real difference (Table 26, Appendix F). The probes were therefore run again with the draws kept, 16 per probe, and seq is now compared paired by draw, with bootstrap bounds over whole draws (Tables 6 and 7; `PLAN-REVISION-3.md`); the draws are part of the deposit, and Tables 6–8 can be rebuilt from them alone (Section 9).

- The surrogate, whose draws are independent of the literal stream's, meets the pre-registered lines (realised gain along the requested update within 10 %, mean difference beyond sampling noise at most 0.10, median variance ratio in $[0.5, 2]$) on 10 of 12 probes, including layers that lose up to 37.5 % of the requested pulse mass to clipping; its variance ratio lies between 0.78 and 1.41. These are point estimates: with 16 draws its bootstrap bounds stay within the lines on 1 of them.

Table 4: Revised main table (repaired simulator): gap to FP-SGD at the same learning rate, mean [95 % $t$-interval] over 5 seeds paired by seed (same initialisation and data order); MNIST and CIFAR in accuracy points, LM in % of bits per byte; P/F/U = PASS/FAIL/UNRESOLVED against the absolute lines 2 pp, 2 pp, 5 %.

| Device | Alg. | MNIST | CIFAR | LM |
|---|---|---|---|---|
| Li-ECRAM | SGD | 6.41 [3.90, 8.93] F | 19.19 [14.94, 23.44] F | 51.64 [31.64, 71.63] F |
| Li-ECRAM | TTv2 | 0.44 [0.24, 0.64] P | 8.65 [6.64, 10.66] F | 8.51 [4.72, 12.30] U |
| Li-ECRAM | c-TTv2 | 0.62 [0.10, 1.14] P | -0.21 [-2.22, 1.80] P | -1.45 [-4.07, 1.18] P |
| Li-ECRAM | AGAD | 0.57 [0.19, 0.94] P | -0.84 [-2.20, 0.53] P | -1.13 [-3.69, 1.44] P |
| MO-ECRAM | SGD | 11.51 [4.88, 18.14] F | 27.24 [20.15, 34.34] F | 73.84 [7.33, 140.35] F |
| MO-ECRAM | TTv2 | 1.93 [1.82, 2.03] U | 7.04 [5.25, 8.83] F | 19.31 [15.06, 23.57] F |
| MO-ECRAM | c-TTv2 | 1.04 [0.79, 1.28] P | 0.33 [-1.60, 2.27] U | 11.33 [7.11, 15.55] F |
| MO-ECRAM | AGAD | 0.80 [0.67, 0.93] P | -0.79 [-2.19, 0.62] P | 7.98 [6.08, 9.87] F |
| Protonic-inspired | SGD | 4.09 [2.29, 5.88] F | 12.66 [8.83, 16.49] F | 35.99 [29.87, 42.10] F |
| Protonic-inspired | TTv2 | 0.57 [0.07, 1.07] P | 6.08 [4.06, 8.10] F | 5.34 [1.89, 8.78] U |
| Protonic-inspired | c-TTv2 | 0.30 [0.09, 0.51] P | -0.37 [-1.97, 1.23] P | -1.53 [-5.07, 2.01] P |
| Protonic-inspired | AGAD | 0.75 [0.45, 1.04] P | -1.34 [-2.41, -0.27] P | -0.04 [-3.84, 3.76] P |
| W/WO$_x$ ECRAM | SGD | 12.06 [6.69, 17.42] F | 33.00 [26.34, 39.67] F | 52.03 [43.34, 60.71] F |
| W/WO$_x$ ECRAM | TTv2 | 1.47 [1.37, 1.56] P | 12.53 [10.59, 14.48] F | 22.48 [18.93, 26.04] F |
| W/WO$_x$ ECRAM | c-TTv2 | 1.52 [1.31, 1.74] P | 10.67 [8.65, 12.69] F | 19.02 [14.49, 23.55] F |
| W/WO$_x$ ECRAM | AGAD | 1.07 [0.71, 1.43] P | 12.72 [11.44, 14.01] F | 21.64 [17.52, 25.76] F |
| Memistor (optimistic) | SGD | 1.60 [0.87, 2.33] U | 6.05 [4.35, 7.75] F | 25.72 [20.59, 30.86] F |
| Memistor (optimistic) | TTv2 | 0.91 [0.64, 1.17] P | 1.39 [0.26, 2.52] U | 3.70 [0.28, 7.12] U |
| Memistor (optimistic) | c-TTv2 | 0.18 [-0.09, 0.46] P | 0.44 [-1.37, 2.25] U | 3.61 [0.14, 7.07] U |
| Memistor (optimistic) | AGAD | 0.62 [0.29, 0.96] P | 0.03 [-1.07, 1.12] P | 3.75 [0.44, 7.07] U |
| Ideal 1000-level | SGD | 0.01 [-0.12, 0.13] P | 6.32 [2.22, 10.42] F | 12.14 [10.28, 14.00] F |
| Ideal 1000-level | TTv2 | 0.83 [0.51, 1.16] P | 4.72 [0.24, 9.20] U | 19.55 [16.07, 23.03] F |
| Ideal 1000-level | c-TTv2 | 0.30 [-0.00, 0.60] P | 18.04 [10.46, 25.61] F | 14.47 [11.99, 16.96] F |
| Ideal 1000-level | AGAD | 0.78 [0.42, 1.14] P | 19.20 [13.21, 25.19] F | 12.02 [9.00, 15.03] F |
| Digital | Adam | 0.03 [-0.18, 0.23] P | -1.07 [-2.65, 0.52] P | -20.19 [-22.39, -17.99] P |

- It fails on the two probes of the most clipped layer: CIFAR MO-ECRAM AGAD layer 4 at update 600 (95.7 % lost, gain +1.5 %, excess 0.15); CIFAR MO-ECRAM AGAD layer 4 at update 2000 (97.3 % lost, gain -4.2 %, excess 0.30). There almost nothing of the requested update survives the clipping, and the surrogate's Poisson counts, which can exceed the slot budget that caps the stream, no longer match it (review 2's limiting case: Poisson(31) exceeds 31 with probability 0.4524).

- Paired by draw, seq meets its registered lines (gain within 5 %, excess at most 0.05, variance ratio in $[0.8, 1.25]$) on 12 of 12 probes. The paired floor, 0.29–0.98 times the independent one, accounts for the raw relative difference (at most 0.224) up to an excess of at most 0.015; gains agree within 1.4 % and variance ratios lie in 0.993–1.029. The bootstrap bounds stay within the lines on 6 probes (LM layer 0 at update 750; LM layer 0 at update 2250; MNIST layer 0 at update 1000; MNIST layer 2 at update 1000; MNIST layer 0 at update 3500; MNIST layer 2 at update 3500). On the others (CIFAR layer 3 at update 600; CIFAR layer 4 at update 600; CIFAR layer 3 at update 2000; CIFAR layer 4 at update 2000; LM layer 9 at update 750; LM layer 9 at update 2250) 16 draws cannot exclude a mean discrepancy of up to 0.286. Seq therefore meets our registered diagnostic thresholds; that is a diagnostic, not a demonstration that it is equivalent to the literal stream.

- Shared lines make the literal stream's cells dependent; the surrogate's independent draws do not reproduce this. Measured by per-pair correlations, which cannot cancel (Table 8), the

Table 5: Revised main table (repaired simulator): gap to digital Adam (its own selected learning rate), mean [95 % $t$-interval] over 5 seeds paired by seed (same initialisation and data order); MNIST and CIFAR in accuracy points, LM in % of bits per byte; P/F/U = PASS/FAIL/UNRESOLVED against the absolute lines 2 pp, 2 pp, 5 %.

| Device | Alg. | MNIST | CIFAR | LM |
|---|---|---|---|---|
| Li-ECRAM | SGD | 6.39 [3.79, 8.98] F | 20.25 [14.51, 26.00] F | 90.33 [60.59, 120.07] F |
| Li-ECRAM | TTv2 | 0.42 [0.21, 0.63] P | 9.72 [8.82, 10.62] F | 35.95 [34.87, 37.02] F |
| Li-ECRAM | c-TTv2 | 0.60 [0.26, 0.93] P | 0.86 [0.17, 1.55] P | 23.49 [22.61, 24.36] F |
| Li-ECRAM | AGAD | 0.54 [0.26, 0.82] P | 0.23 [-0.38, 0.84] P | 23.89 [22.95, 24.82] F |
| MO-ECRAM | SGD | 11.48 [4.73, 18.23] F | 28.31 [20.59, 36.03] F | 117.64 [35.42, 199.87] F |
| MO-ECRAM | TTv2 | 1.90 [1.77, 2.04] U | 8.11 [7.85, 8.38] F | 49.48 [47.97, 50.98] F |
| MO-ECRAM | c-TTv2 | 1.01 [0.96, 1.06] P | 1.40 [0.28, 2.53] U | 39.47 [37.39, 41.55] F |
| MO-ECRAM | AGAD | 0.78 [0.60, 0.96] P | 0.28 [-0.30, 0.86] P | 35.33 [31.48, 39.19] F |
| Protonic-inspired | SGD | 4.06 [2.18, 5.94] F | 13.73 [10.41, 17.05] F | 70.37 [65.57, 75.16] F |
| Protonic-inspired | TTv2 | 0.54 [0.23, 0.86] P | 7.15 [5.27, 9.04] F | 31.98 [30.89, 33.06] F |
| Protonic-inspired | c-TTv2 | 0.27 [0.16, 0.38] P | 0.70 [-1.68, 3.07] U | 23.36 [22.15, 24.58] F |
| Protonic-inspired | AGAD | 0.72 [0.58, 0.86] P | -0.27 [-1.30, 0.75] P | 25.23 [23.43, 27.03] F |
| W/WO$_x$ ECRAM | SGD | 12.03 [6.56, 17.50] F | 34.07 [28.80, 39.34] F | 90.44 [82.88, 97.99] F |
| W/WO$_x$ ECRAM | TTv2 | 1.44 [1.23, 1.66] P | 13.60 [11.76, 15.44] F | 53.46 [52.68, 54.24] F |
| W/WO$_x$ ECRAM | c-TTv2 | 1.50 [1.39, 1.60] P | 11.74 [10.96, 12.52] F | 49.10 [47.30, 50.90] F |
| W/WO$_x$ ECRAM | AGAD | 1.04 [0.66, 1.42] P | 13.79 [12.70, 14.88] F | 52.40 [51.04, 53.75] F |
| Memistor (optimistic) | SGD | 1.58 [0.75, 2.41] U | 7.12 [4.95, 9.30] F | 57.52 [53.05, 61.99] F |
| Memistor (optimistic) | TTv2 | 0.88 [0.57, 1.19] P | 2.46 [0.65, 4.28] U | 29.93 [28.82, 31.03] F |
| Memistor (optimistic) | c-TTv2 | 0.16 [0.01, 0.30] P | 1.51 [-0.47, 3.49] U | 29.80 [28.39, 31.22] F |
| Memistor (optimistic) | AGAD | 0.60 [0.35, 0.84] P | 1.09 [-0.24, 2.43] U | 29.99 [28.42, 31.56] F |
| Ideal 1000-level | SGD | -0.02 [-0.22, 0.18] P | 7.39 [3.31, 11.46] F | 40.53 [38.35, 42.71] F |
| Ideal 1000-level | TTv2 | 0.81 [0.54, 1.07] P | 5.79 [0.66, 10.91] U | 49.79 [48.21, 51.37] F |
| Ideal 1000-level | c-TTv2 | 0.27 [0.10, 0.44] P | 19.10 [10.33, 27.87] F | 43.45 [41.48, 45.41] F |
| Ideal 1000-level | AGAD | 0.75 [0.47, 1.03] P | 20.27 [15.32, 25.22] F | 40.36 [38.70, 42.01] F |

root-mean-square correlation among cells sharing a line exceeds the surrogate's sampling null by up to 0.345 (mean correlation aligned with the expected update's signs up to 0.208), by more than 0.05 on 6 of 12 probes (LM layer 9 at update 750; LM layer 9 at update 2250; MNIST layer 0 at update 1000; MNIST layer 2 at update 1000; MNIST layer 0 at update 3500; MNIST layer 2 at update 3500). The statistic of version 3, a pooled, signed normalised covariance, stays within 0.020 in absolute value on the same draws, because positive and negative terms cancel and unequal cell variances are pooled. The cross-cell dependence is therefore not small everywhere; the per-cell lines above do not test it, and whether it matters for training is what A2 measures.

The probes draw from a generator of their own. On the pod (version 3) the MNIST and CIFAR probe runs reproduced their main-table records exactly; on the desk (Quadro P2200) 0 of 3 do, and none shares its initialisation with its version-3 run (the MNIST run not its example order either): the simulator's generators run on the GPU, and the CUDA random kernels assign draws to threads by the GPU's multiprocessor count. The desk's probe runs are therefore fresh training realisations, and the probed states are not version 3's; the comparison within each probe is unaffected, since all three updates start from the same saved state. LM training is not bitwise reproducible even on one card type: configurations repeated exactly in the records differ by up to 0.31 % in bits per byte (CIFAR 0.00 %).

**Training level (A2).** The same selections, learning rates and seeds as the main table were trained with the per-sample stream (MNIST: every analog layer, three devices × SGD, c-TTv2, AGAD; CIFAR: the two fully connected layers, whose rows are one per image, for Li-ECRAM and protonic-inspired c-TTv2 and MO-ECRAM AGAD; the convolutional layers keep the surrogate), seeds 0–2, and compared seed by seed with the main table's surrogate runs; both arms are labelled at the same three seeds (Table 10). The pre-registered rule calls the change material if any cell's mean gap moves by more than half the line (1 pp) or two cells whose gaps differ by more than 1 pp swap order.

- By that rule the result is **material**, through one cell: MNIST MO-ECRAM SGD, whose gap moves from 14.10 to 12.68 pp (difference -1.42 pp, 95 % interval [-4.84, 2.00]). At three seeds both of its arms are UNRESOLVED (surrogate [0.04, 28.15], stream [1.27, 24.09] pp). The rerun flagged in version 3 was done as matched triples on one GPU (experiment A2-M, Table 9; `PLAN-REVISION-4.md`): on the desk's Quadro P2200, each of seeds 0–4 trained the FP-SGD control, the surrogate and the stream from one initialisation and one example order (equal hashes, checked; the analysis refuses a seed without them). At these 5 matched seeds the stream's gap is 8.73 pp [6.82, 10.65], FAIL, against 8.71 pp [7.02, 10.40], FAIL, with the surrogate; the paired change is 0.02 pp [-0.47, 0.52] (not material; its interval lies within ±1 pp). The FAIL label holds under either update. The pod's runs of seeds 0–2 are other random realisations of the same seeds, not a rounding variant of them: the simulator's generators run on the GPU, and the CUDA random kernels assign draws to threads by the GPU's multiprocessor count, so another GPU model draws another initialisation and example order (as predicted, the hashes differ from both the pod's and the earlier desk runs'). Per seed the two realisations' gaps differ by up to 9.97 pp, and at the pod's three matched seeds the change was -1.42 pp [-4.84, 2.00]: this cell's gap depends strongly on the realisation, and the update's effect on it is measured only by the paired comparisons within each matched set. Version 4's five-seed figures for this cell mixed matched and unmatched pairs and are withdrawn (Appendix F).

- One label changes at the same three seeds: CIFAR Li-ECRAM c-TTv2 (PASS with the surrogate, UNRESOLVED with the stream). Its mean gap moves by only 0.52 pp, but the stream's interval ([-6.44, 5.20] pp) is wider and includes the line. This is not a failure, but the label is not preserved, and the difference interval ([-3.26, 4.29] pp) does not bound the change within ±1 pp.

- 2 pairs of cells reverse their mean order (MNIST MO-ECRAM AGAD and MNIST Protonic-inspired AGAD (separation -0.02 to 0.12 pp); CIFAR Li-ECRAM c-TTv2 and CIFAR Protonic-inspired c-TTv2 (separation -0.29 to 0.84 pp)); none was separated by more than 1 pp under the surrogate, so there is no material rank reversal under the registered rule, but the ranking as a whole is not preserved. No Tiki-Taka-family cell's mean moves by more than 0.61 pp (MNIST at most 1.42 pp including the SGD cell; CIFAR at most 0.61 pp, with intervals of several pp at three seeds).

Within these bounds (MNIST on every layer, CIFAR on its fully connected layers, the LM at update level only, three seeds), the registered result is one material mean change, in a failing SGD cell whose FAIL label holds at five matched seeds on one GPU under either update; no material rank reversal; no Tiki-Taka-family mean shift above 0.61 pp; and one label that loses PASS. This is not a demonstration that the surrogate and the stream train equivalently, and the per-sample stream is itself an approximation of the literal one. The surrogate is not validated where clipping

Table 6: Update fidelity of the aggregate surrogate against the literal pulse stream (experiment A1-P; desk rerun of A1 with the draws kept, 16 draws per probe; Li = Li-ECRAM, MO = MO-ECRAM, Prot. = protonic-inspired). Lost = requested pulse mass removed by the probability clipping (%). The surrogate's draws are independent of the stream's, so the independent sampling floor applies: gain difference (%, bootstrap 95 % interval over draws), excess relative mean difference beyond the floor (bootstrap 95th percentile), median variance ratio, and whether the point estimates meet the registered lines (10 %, 0.10, [0.5, 2]). "(95)": the bootstrap 95th percentile of the excess and the bootstrap 95 % gain interval are also within their lines; it does not cover the variance ratio (a point estimate) and is not a joint 95 % guarantee for every line.

| Case | L. | Upd. | Lost | $\Delta g$ % | excess | var. | meets |
|---|---|---|---|---|---|---|---|
| CIFAR MO AGAD | 3 | 600 | 36.2 | 1.4 [1.2, 1.6] | 0.06 (0.17) | 1.14 | yes |
| CIFAR MO AGAD | 4 | 600 | 95.7 | 1.5 [0.5, 2.6] | 0.15 (0.16) | 1.26 | no |
| CIFAR MO AGAD | 3 | 2000 | 26.2 | 0.5 [-0.3, 1.3] | 0.03 (0.31) | 1.07 | yes |
| CIFAR MO AGAD | 4 | 2000 | 97.3 | -4.2 [-7.8, -0.2] | 0.30 (0.32) | 1.27 | no |
| LM Prot. c-TTv2 | 0 | 750 | 6.1 | 1.3 [0.8, 1.8] | 0.03 (0.14) | 1.01 | yes |
| LM Prot. c-TTv2 | 9 | 750 | 0.0 | -0.5 [-2.4, 1.8] | 0.03 (0.32) | 0.86 | yes |
| LM Prot. c-TTv2 | 0 | 2250 | 7.0 | -0.6 [-1.0, -0.1] | 0.04 (0.13) | 1.05 | yes |
| LM Prot. c-TTv2 | 9 | 2250 | 0.0 | 0.3 [-1.8, 2.3] | 0.04 (0.32) | 0.78 | yes |
| MNIST Li c-TTv2 | 0 | 1000 | 16.0 | -0.4 [-0.8, 0.1] | 0.00 (0.14) | 1.06 | yes |
| MNIST Li c-TTv2 | 2 | 1000 | 4.6 | -1.1 [-2.6, 0.6] | 0.02 (0.10) | 1.19 | yes |
| MNIST Li c-TTv2 | 0 | 3500 | 37.5 | 0.6 [0.0, 1.1] | 0.00 (0.15) | 1.11 | yes |
| MNIST Li c-TTv2 | 2 | 3500 | 5.7 | 0.7 [-0.5, 2.1] | 0.02 (0.09) | 1.41 | yes (95) |

removes nearly all of the update (the CIFAR output layer above), for the convolutional layers, or for the LM's training trajectory, and the main results remain comparisons within this aggregate-update simulator.

### 4.4 Width

An earlier lane of this project saw the device gap grow with model width. Here every arm (protonic-inspired and Li-ECRAM, each with TTv2 and c-TTv2) gets its own learning rate at every width by the edge rule on validation, and then five test seeds per width; FP-SGD uses its own selected rate (Table 11; the rates are in Table 14). As in the rest of the paper, each layer's weight scale $\alpha$ is set from its initialisation bound, so it falls as $1/\sqrt{\text{fan-in}}$ when the width raises a layer's fan-in (layers with a fixed fan-in keep theirs); this is one mapping policy, not every fixed physical mapping.

**The test.** The endpoint contrast $D$ (the mean gap at the widest minus the narrowest width) is tested for equivalence within the gap lines (TOST at 5 %). The original paper's "FLAT" label is withdrawn: it did not test equivalence and covered large non-monotonic changes. The pre-registered contrast is Welch's, which treats the two widths' seed samples as independent. They are not: every arm of a seed shares the logical initialisation and the data order across widths. Beside the registered result we therefore report the contrast paired by seed, with the same decision order, as a sensitivity analysis added after the second review (deviation R4). Both test the endpoints only; neither covers the middle width.

**CIFAR ($c = 16$, 32, 64).** 1 arm is equivalent within ±2 pp (Li-ECRAM c-TTv2, $D = -0.34$ pp, 90 % interval [-1.34, 0.66]); for 1 arm there is no evidence of growth beyond the margin (protonic-inspired TTv2); 2 arms are inconclusive (Li-ECRAM TTv2, $D = 1.91$ pp; protonic-inspired c-TTv2,

Table 7: Update fidelity of the per-sample stream (seq) against the literal pulse stream, paired by draw (shared line firings; experiment A1-P, the probes of Table 6): gain difference (%, bootstrap 95 % interval), raw relative mean difference, independent / paired sampling floor, excess beyond the PAIRED floor (bootstrap 95th percentile), median variance ratio, and whether the point estimates meet the registered lines (5 %, 0.05, [0.8, 1.25]). "(95)": the bootstrap 95th percentile of the excess and the bootstrap 95 % gain interval are also within their lines; it does not cover the variance ratio (a point estimate) and is not a joint 95 % guarantee for every line.

| Case | L. | Upd. | $\Delta g$ % | raw | floor ind./pair. | excess | var. | meets |
|---|---|---|---|---|---|---|---|---|
| CIFAR MO AGAD | 3 | 600 | 0.1 [-0.1, 0.2] | 0.114 | 0.121 / 0.114 | 0.011 (0.144) | 1.00 | yes |
| CIFAR MO AGAD | 4 | 600 | -0.9 [-1.5, -0.3] | 0.050 | 0.050 / 0.050 | 0.005 (0.064) | 1.03 | yes |
| CIFAR MO AGAD | 3 | 2000 | -0.2 [-0.6, 0.1] | 0.224 | 0.241 / 0.224 | 0.015 (0.286) | 1.00 | yes |
| CIFAR MO AGAD | 4 | 2000 | -1.4 [-5.5, 3.5] | 0.072 | 0.074 / 0.073 | 0.000 (0.094) | 1.00 | yes |
| LM Prot. c-TTv2 | 0 | 750 | -0.0 [-0.2, 0.2] | 0.032 | 0.106 / 0.032 | 0.000 (0.041) | 1.00 | yes (95) |
| LM Prot. c-TTv2 | 9 | 750 | 0.0 [-0.3, 0.3] | 0.075 | 0.258 / 0.075 | 0.000 (0.095) | 1.00 | yes |
| LM Prot. c-TTv2 | 0 | 2250 | 0.1 [-0.0, 0.3] | 0.029 | 0.093 / 0.028 | 0.004 (0.037) | 1.00 | yes (95) |
| LM Prot. c-TTv2 | 9 | 2250 | 0.2 [-0.2, 0.5] | 0.073 | 0.249 / 0.073 | 0.007 (0.093) | 1.00 | yes |
| MNIST Li c-TTv2 | 0 | 1000 | -0.0 [-0.0, 0.0] | 0.036 | 0.109 / 0.036 | 0.001 (0.045) | 1.00 | yes (95) |
| MNIST Li c-TTv2 | 2 | 1000 | 0.1 [-0.1, 0.2] | 0.027 | 0.074 / 0.027 | 0.000 (0.035) | 1.02 | yes (95) |
| MNIST Li c-TTv2 | 0 | 3500 | -0.0 [-0.1, 0.0] | 0.036 | 0.113 / 0.036 | 0.003 (0.046) | 1.00 | yes (95) |
| MNIST Li c-TTv2 | 2 | 3500 | 0.1 [-0.2, 0.3] | 0.024 | 0.063 / 0.025 | 0.000 (0.031) | 0.99 | yes (95) |

$D = 1.76$ pp). Five seeds are not enough to settle a 2 pp margin on this subset. Paired by seed, the Li-ECRAM c-TTv2 equivalence survives (90 % interval [-1.79, 1.11]), and protonic-inspired c-TTv2 *grows* ([0.15, 3.38]): the growth is detected, but its size against the margin is unresolved: the interval includes 2 pp, so growth beyond the margin is not excluded. Both of its endpoint cells still pass against their own FP-SGD controls; a gap that moves from more negative to less negative can grow while both cells stay acceptable. Li-ECRAM TTv2 stays inconclusive, so growth beyond the margin is not excluded for it either.

**The LM ($d = 64$, 128, 256).** All 4 arms have a gap that shrinks with width (for example Li-ECRAM c-TTv2 from 1.21 % to -12.90 %). This is not a favourable scaling law: FP-SGD itself gets worse at the largest width under the fixed budget (2.962 bits per byte at $d = 128$, 3.117 at $d = 256$), so a gap that turns negative partly measures a weaker baseline. Converged or quality-matched baselines would be needed to speak of scaling.

**What can be said.** Across a factor of four in width, after per-arm learning-rate selection, no arm's gap grows detectably beyond the margin. Endpoint equivalence is established for one CIFAR arm only (Li-ECRAM c-TTv2, by both contrasts); the paired analysis detects growth for one other CIFAR arm (protonic-inspired c-TTv2), which the registered Welch test does not, without resolving its size against the margin; for two arms (protonic-inspired c-TTv2 and Li-ECRAM TTv2) growth beyond the margin is not excluded. On the LM the apparent shrinkage reflects the FP-SGD baseline at a fixed budget. Width dependence is unresolved by this study.

### 4.5 Sensitivity against absolute lines

Around the protonic-inspired base we move one device property at a time (levels, step asymmetry, saturation $\gamma$, write noise, reference error, retention) at the base sizes (CIFAR $c = 32$, LM $d = 128$), three seeds paired with FP-SGD of the same seed, at FP-SGD's learning rate. Every grid point is

Table 8: Cross-cell dependence of the literal stream's draws (experiment A1-P), among cells sharing a row / a column, from per-pair Pearson correlations over all pairs (they cannot cancel): root-mean-square correlation of the literal stream and of the surrogate (whose cells are independent, so its value is the sampling null at the same number of draws), the excess RMS beyond that null, the mean correlation aligned with the signs of the expected update, the largest per-line RMS, and the version-3 statistic (a pooled, signed normalised covariance, which can cancel).

| Case | L. | Upd. | RMS slot | RMS sur. (null) | excess RMS | aligned mean | max line RMS | v3 pooled |
|---|---|---|---|---|---|---|---|---|
| CIFAR MO AGAD | 3 | 600 | 0.258 / 0.261 | 0.258 / 0.258 | 0.009 / 0.041 | 0.005 / 0.010 | 0.262 / 1.000 | 0.000 / -0.000 |
| CIFAR MO AGAD | 4 | 600 | 0.256 / 0.261 | 0.259 / 0.260 | 0.000 / 0.026 | 0.007 / -0.004 | 0.259 / 0.645 | 0.000 / -0.008 |
| CIFAR MO AGAD | 3 | 2000 | 0.260 / 0.260 | 0.258 / 0.258 | 0.032 / 0.026 | 0.007 / 0.004 | 0.283 / 0.399 | -0.000 / -0.000 |
| CIFAR MO AGAD | 4 | 2000 | 0.259 / 0.260 | 0.259 / 0.255 | 0.015 / 0.048 | 0.006 / -0.003 | 0.265 / 0.315 | 0.000 / 0.001 |
| LM Prot. c-TTv2 | 0 | 750 | 0.259 / 0.258 | 0.258 / 0.258 | 0.015 / 0.008 | -0.000 / 0.000 | 0.266 / 0.264 | 0.000 / 0.000 |
| LM Prot. c-TTv2 | 9 | 750 | 0.299 / 0.258 | 0.258 / 0.258 | 0.152 / 0.007 | 0.026 / -0.000 | 1.000 / 0.264 | -0.000 / -0.001 |
| LM Prot. c-TTv2 | 0 | 2250 | 0.258 / 0.258 | 0.258 / 0.258 | 0.000 / 0.000 | -0.000 / 0.000 | 0.265 / 0.262 | 0.000 / -0.000 |
| LM Prot. c-TTv2 | 9 | 2250 | 0.315 / 0.258 | 0.258 / 0.258 | 0.180 / 0.003 | 0.025 / -0.000 | 1.000 / 0.266 | 0.001 / -0.001 |
| MNIST Li c-TTv2 | 0 | 1000 | 0.335 / 0.262 | 0.258 / 0.258 | 0.213 / 0.044 | 0.054 / 0.010 | 0.454 / 0.589 | -0.000 / -0.000 |
| MNIST Li c-TTv2 | 2 | 1000 | 0.399 / 0.262 | 0.258 / 0.260 | 0.305 / 0.031 | 0.103 / 0.014 | 0.494 / 0.348 | -0.000 / -0.020 |
| MNIST Li c-TTv2 | 0 | 3500 | 0.385 / 0.259 | 0.258 / 0.258 | 0.286 / 0.025 | 0.100 / 0.004 | 0.478 / 0.322 | -0.001 / -0.000 |
| MNIST Li c-TTv2 | 2 | 3500 | 0.431 / 0.264 | 0.258 / 0.258 | 0.345 / 0.057 | 0.208 / 0.003 | 0.522 / 0.321 | 0.006 / -0.003 |

Table 9: Experiment A2-M: MNIST MO-ECRAM SGD as matched triples on one GPU (Quadro P2200): per seed, the FP-SGD control's, the surrogate's and the per-sample stream's test accuracy (%) and the two gaps to the control (pp); every seed's three runs share their initialisation and example order (equal hashes, checked). Right: the same seeds' gaps from the pod's matched triples (RTX A4000), a different random realisation of each seed.

| | P2200 | | | | | pod (A4000) | |
|---|---|---|---|---|---|---|---|
| Seed | FP-SGD | sur. | seq | gap sur. | gap seq | gap sur. | gap seq |
| 0 | 97.90 | 89.83 | 90.02 | 8.07 | 7.88 | 16.84 | 15.69 |
| 1 | 97.60 | 87.86 | 87.30 | 9.74 | 10.30 | 7.59 | 7.39 |
| 2 | 97.68 | 89.79 | 90.26 | 7.89 | 7.42 | 17.86 | 14.95 |
| 3 | 97.74 | 87.20 | 87.22 | 10.54 | 10.52 | – | – |
| 4 | 97.75 | 90.44 | 90.20 | 7.31 | 7.55 | – | – |
| mean [95 %] | | | | 8.71 [7.02, 10.40] | 8.73 [6.82, 10.65] | | |

labelled against the *absolute* line (2 pp, 5 %), never against the base device's own gap (Table 12). The asymmetry sweep is run twice: as before, where the symmetry point $u^* = a/(2\gamma)$ moves toward the upper bound and the positive weight range shrinks ($a = 0.4$ at $\gamma = 0.2$ puts $u^*$ on the bound); and with each device's range re-centred on its symmetry point, so that only the up/down dynamics change.

**The pre-registered sheet (TTv2) establishes no threshold.** Its base point is itself UNRESOLVED on both tasks (6.35 pp on CIFAR, 5.15 % on the LM), so by the pre-registered rule every axis is "not established". It still shows the confound the review pointed out: at $a = 0.4$ the gap is 42.92 pp on CIFAR when the symmetry point moves onto the bound, but 6.20 pp when it is held at mid-range. The earlier "asymmetry limit" was mostly lost weight range.

**The chopped-TTv2 sheet (added after the main table, deviation R3).** On CIFAR the base passes (-0.85 pp, PASS), and the sheet establishes: step asymmetry up to 0.2 with the symme-

Table 10: Training fidelity (experiment A2), seeds 0–2 on both arms: mean gap to FP-SGD (pp) with its 95 % interval and label against the 2 pp line (P pass, U unresolved, F fail), with the aggregate surrogate (E3's records) and with the per-sample pulse stream (MNIST: every analog layer; CIFAR: the two fully connected layers); the paired difference with its 95 % interval; material by the registered rule (|mean| > 1 pp). The labels are at $n = 3$; the main table's labels are at $n = 5$.

| Task | Cell | surrogate | stream | difference | material |
|---|---|---|---|---|---|
| CIFAR | Li-ECRAM c-TTv2 | -1.14 [-3.89, 1.62] P | -0.62 [-6.44, 5.20] U | 0.52 [-3.26, 4.29] | no |
| CIFAR | MO-ECRAM AGAD | -1.40 [-3.22, 0.41] P | -1.82 [-3.76, 0.13] P | -0.41 [-1.99, 1.16] | no |
| CIFAR | Protonic-inspired c-TTv2 | -0.85 [-3.33, 1.64] P | -1.46 [-3.94, 1.02] P | -0.61 [-2.86, 1.64] | no |
| MNIST | Li-ECRAM AGAD | 0.65 [-0.26, 1.56] P | 0.49 [0.08, 0.90] P | -0.16 [-0.69, 0.36] | no |
| MNIST | Li-ECRAM c-TTv2 | 0.68 [-0.77, 2.12] U | 0.74 [-1.70, 3.18] U | 0.07 [-0.95, 1.09] | no |
| MNIST | Li-ECRAM SGD | 6.99 [0.52, 13.46] U | 6.90 [1.03, 12.76] U | -0.09 [-0.73, 0.54] | no |
| MNIST | MO-ECRAM AGAD | 0.77 [0.48, 1.06] P | 0.89 [0.70, 1.07] P | 0.11 [-0.07, 0.30] | no |
| MNIST | MO-ECRAM c-TTv2 | 1.08 [0.44, 1.72] P | 1.05 [0.43, 1.66] P | -0.04 [-0.23, 0.16] | no |
| MNIST | MO-ECRAM SGD | 14.10 [0.04, 28.15] U | 12.68 [1.27, 24.09] U | -1.42 [-4.84, 2.00] | yes |
| MNIST | Protonic-inspired AGAD | 0.79 [0.15, 1.43] P | 0.77 [0.17, 1.37] P | -0.02 [-0.39, 0.35] | no |
| MNIST | Protonic-inspired c-TTv2 | 0.29 [-0.30, 0.88] P | 0.21 [-0.71, 1.13] P | -0.08 [-0.44, 0.27] | no |
| MNIST | Protonic-inspired SGD | 4.58 [0.15, 9.00] U | 4.69 [0.01, 9.38] U | 0.12 [-0.33, 0.56] | no |

Table 11: Revised width study (Li = Li-ECRAM, Prot. = protonic-inspired): mean gap at the narrow / mid / wide width (5 seeds, each arm at its own selected learning rate; CIFAR $c = 16/32/64$ in pp, LM $d = 64/128/256$ in %), and the endpoint contrast $D$ (widest minus narrowest) with its 90 % interval and verdict, by the pre-registered Welch test and, added after review 2 (deviation R4), paired by seed. Verdicts (margins 2 pp, 5 %): EQ = equivalent at the endpoints, GR = grows, SH = shrinks, NG = no evidence of growth beyond the margin, INC = inconclusive. An endpoint test says nothing about the middle width.

| Task | Arm | gap narrow / mid / wide | Welch $D$ [90 %] | paired $D$ [90 %] |
|---|---|---|---|---|
| CIFAR | Li c-TTv2 | -0.36 / 0.66 / -0.71 | -0.34 [-1.34, 0.66] EQ | -0.34 [-1.79, 1.11] EQ |
| CIFAR | Li TTv2 | 3.29 / 4.12 / 5.20 | 1.91 [-0.37, 4.19] INC | 1.91 [-1.53, 5.35] INC |
| CIFAR | Prot. c-TTv2 | -2.57 / -1.55 / -0.81 | 1.76 [-0.02, 3.55] INC | 1.76 [0.15, 3.38] GR |
| CIFAR | Prot. TTv2 | 1.38 / 2.71 / 0.43 | -0.95 [-2.53, 0.62] NG | -0.95 [-3.36, 1.46] NG |
| LM | Li c-TTv2 | 1.21 / -4.69 / -12.90 | -14.12 [-15.38, -12.86] SH | -14.12 [-15.30, -12.94] SH |
| LM | Li TTv2 | 9.85 / 13.29 / -3.07 | -12.92 [-14.59, -11.25] SH | -12.92 [-14.99, -10.85] SH |
| LM | Prot. c-TTv2 | -5.79 / -6.36 / -12.69 | -6.90 [-8.10, -5.70] SH | -6.90 [-7.76, -6.04] SH |
| LM | Prot. TTv2 | 7.95 / 0.39 / -6.49 | -14.44 [-19.30, -9.57] SH | -14.44 [-19.31, -9.57] SH |

try point moving (at 0.4 the gap jumps to 42.39 pp, when $u^*$ reaches the bound), and reference error up to 0.05 of the half range. With the symmetry point held at mid-range, version 2 borrowed the ordinary base (whose range is $[-1.125, +0.875]$) as this sweep's anchor and omitted its zero point (deviation R5); both are now run with each device's range re-centred (range $[-1, 1]$, pre-registered after the second review). The recentred base passes (-0.41 pp) and so does $a = 0$ (-1.43 pp), so the held-symmetry threshold 0.05 is now established by its own runs; it differs from the unrecentred base by 0.44 pp (95 % interval [-2.66, 3.54]). No point fails: up to $a = 0.4$ the mean gap stays at -0.53 pp and the intervals merely include the line. Levels, saturation, write noise and finite retention are not established: neighbouring grid values are UNRESOLVED with three seeds, while the clear failures are no saturation ($\gamma = 0$: 21.14 pp), strong saturation ($\gamma = 1$: 8.01 pp), reference error 0.2 (9.31 pp) and retention of $10^3$ updates or less. On the LM the base is UNRESOLVED with three seeds (-1.84 %; it passed with five seeds in the main table), so no LM threshold is established.

Table 12: Revised sensitivity sheet (protonic-inspired base; TTv2 = the pre-registered S-ALGO, c-TTv2 added after the main table was seen, deviation R3; 3 seeds paired with FP-SGD): mean gap per grid value with its label against the ABSOLUTE line (P/F/U), and the weakest value that passes with every value toward the best end also passing (– = not established).

| Alg., task | Axis | mean gap per value | threshold |
|---|---|---|---|
| c-TTv2, CIFAR | levels | 5000.0: 2.49U; 2000.0: -2.43P; 1000: -0.85P; 500.0: -1.57P; 200.0: 1.38U; 100.0: 1.68U | – |
| c-TTv2, CIFAR | asym. $a$ (SP moves) | 0.0: -1.43P; 0.05: -0.85P; 0.1: -1.64P; 0.2: -1.16P; 0.4: 42.39F | 0.2 |
| c-TTv2, CIFAR | asym. $a$ (SP fixed) | 0.0: -1.43P; 0.05: -0.41P; 0.1: -0.82U; 0.2: -1.38U; 0.4: -0.53U | 0.05 |
| c-TTv2, CIFAR | nonlin. $\gamma$ | 0.0: 21.14F; 0.2: -0.85P; 0.5: 1.88U; 1.0: 8.01F; 2.0: 14.77F | – |
| c-TTv2, CIFAR | noise $c_{2c}$ | 0.0: -0.64U; 0.3: -0.85P; 1.0: -1.34U; 2.0: 0.61U; 4.0: 5.37U | – |
| c-TTv2, CIFAR | ref. err. $\sigma_r$ | 0.0: -1.20P; 0.02: -0.85P; 0.05: -1.29P; 0.1: 2.89U; 0.2: 9.31F; 0.5: 34.63F | 0.05 |
| c-TTv2, CIFAR | retention $\tau$ | inf: -0.85P; 100000.0: -1.17U; 10000.0: 3.08U; 1000.0: 33.02F; 100.0: 44.03F | inf |
| c-TTv2, LM | levels | 5000.0: 2.01U; 2000.0: -2.63U; 1000: -1.84U; 500.0: -0.45U; 200.0: 1.58U; 100.0: 6.54U | – |
| c-TTv2, LM | asym. $a$ (SP moves) | 0.0: -1.68U; 0.05: -1.84U; 0.1: -1.54U; 0.2: -1.85U; 0.4: 37.71U | – |
| c-TTv2, LM | asym. $a$ (SP fixed) | 0.0: -1.62U; 0.05: -2.16U; 0.1: -1.69U; 0.2: -1.70U; 0.4: -1.68U | – |
| c-TTv2, LM | nonlin. $\gamma$ | 0.0: 22.71F; 0.2: -1.84U; 0.5: -0.39U; 1.0: 4.46U; 2.0: 27.27U | – |
| c-TTv2, LM | noise $c_{2c}$ | 0.0: -1.51U; 0.3: -1.84U; 1.0: -0.21U; 2.0: 4.91U; 4.0: 19.04F | – |
| c-TTv2, LM | ref. err. $\sigma_r$ | 0.0: -2.42U; 0.02: -1.84U; 0.05: 1.45U; 0.1: 8.23U; 0.2: 17.53F; 0.5: 30.83F | – |
| c-TTv2, LM | retention $\tau$ | inf: -1.84U; 100000.0: -1.75U; 10000.0: -3.47U; 1000.0: 28.50F; 100.0: 64.34F | – |
| TTv2, CIFAR | levels | 5000.0: 0.78U; 2000.0: 1.67U; 1000: 6.35U; 500.0: 4.43U; 200.0: 3.13U; 100.0: 3.12U | – |
| TTv2, CIFAR | asym. $a$ (SP moves) | 0.0: 5.92U; 0.05: 6.35U; 0.1: 5.95F; 0.2: 6.39F; 0.4: 42.92F | – |
| TTv2, CIFAR | asym. $a$ (SP fixed) | 0.0: 5.92U; 0.05: 6.02U; 0.1: 6.12U; 0.2: 4.60U; 0.4: 6.20U | – |
| TTv2, CIFAR | nonlin. $\gamma$ | 0.0: 27.43F; 0.2: 6.35U; 0.5: 9.00F; 1.0: 8.22F; 2.0: 12.20F | – |
| TTv2, CIFAR | noise $c_{2c}$ | 0.0: 5.79U; 0.3: 6.35U; 1.0: 6.19U; 2.0: 7.18U; 4.0: 9.40F | – |
| TTv2, CIFAR | ref. err. $\sigma_r$ | 0.0: 5.08U; 0.02: 6.35U; 0.05: 5.35U; 0.1: 9.94F; 0.2: 18.74F; 0.5: 30.32F | – |
| TTv2, CIFAR | retention $\tau$ | inf: 6.35U; 100000.0: 4.49U; 10000.0: 5.18U; 1000.0: 40.83F; 100.0: 47.25F | – |
| TTv2, LM | levels | 5000.0: 7.33U; 2000.0: 4.67U; 1000: 5.15U; 500.0: 5.87U; 200.0: 7.38U; 100.0: 9.64U | – |
| TTv2, LM | asym. $a$ (SP moves) | 0.0: 5.26U; 0.05: 5.15U; 0.1: 5.21U; 0.2: 5.00U; 0.4: 10.38U | – |
| TTv2, LM | asym. $a$ (SP fixed) | 0.0: 5.25U; 0.05: 5.15U; 0.1: 5.28U; 0.2: 5.29U; 0.4: 5.75U | – |
| TTv2, LM | nonlin. $\gamma$ | 0.0: 32.24F; 0.2: 5.15U; 0.5: 11.13U; 1.0: 18.03F; 2.0: 25.84F | – |
| TTv2, LM | noise $c_{2c}$ | 0.0: 5.14U; 0.3: 5.15U; 1.0: 5.56U; 2.0: 7.85U; 4.0: 15.03U | – |
| TTv2, LM | ref. err. $\sigma_r$ | 0.0: 4.59U; 0.02: 5.15U; 0.05: 7.27U; 0.1: 11.43U; 0.2: 19.66F; 0.5: 31.81F | – |
| TTv2, LM | retention $\tau$ | inf: 5.15U; 100000.0: 5.09U; 10000.0: 3.74U; 1000.0: 30.34F; 100.0: 71.47F | – |

**How to read the sheet.** These are one-at-a-time ranges around one base device, for one algorithm setting, at FP-SGD's learning rate, with three seeds; they are exploratory, not fabrication tolerances. Combined deviations, other algorithm settings and independent device populations are not covered. A near-zero saturation fails here because, with these settings, the accumulator's decay comes from the device (Section 4.2); it is not a general requirement for training. The rule is deliberately stringent: an UNRESOLVED point at the best end of an axis blocks a threshold even where intermediate points pass. "Not established" therefore does not mean that no grid value passed (on CIFAR with c-TTv2 the 500, 1000 and 2000 level points pass, -1.57, -0.85 and -2.43 pp, while 5000 levels give 2.49 pp, UNRESOLVED), and it does not show that more levels harm hardware training in general.

# 5 A conditional energy projection for the weight-matrix operations

This section is an accounting scenario, not an energy measurement of a training system. It asks how much energy the weight-matrix operations of a training step would take if the analog reads ran at the efficiency of a published analog-AI *inference* chip and the updates cost the published or projected pulse energies, and it compares that with an H100.

**Assumptions, stated first.**

1. **Different read assumptions for accuracy and energy.** The accuracies of Section 4.2 are simulated with ideal fp32 reads; the energy prices finite-precision analog reads. We have not shown that one operating point gives both the reported accuracy and the assumed read efficiency.

2. **The read efficiency is borrowed from a phase-change inference chip** [36]: up to 12.4 TOPS/W sustained, measured on one chip, and 6.94 TOPS/W, the authors' *projected* full-system figure for their five-chip system: the chips were not integrated with digital processing, whose time and energy the authors estimate from their architecture model, and external components such as system buses and voltage regulators are excluded. We report both. Neither is an ECRAM training array, which also needs backward reads, repeated writes, references and Tiki-Taka state.

3. **The boundary is the weight matrices.** Per weight and per training row: the forward and backward reads (two MACs) and the update (device pulses, pulse lines and pulse generation, and the TT family's column reads and digital buffer). Attention products, nonlinearities, normalisation, activation storage, data movement, control, reference maintenance and static power are not modelled on either side, and the digital exclusions do not cancel the analog ones.

4. **The GPU side** is H100 board power over its dense BF16 peak [43] at the utilisation of a very large training run, 38–43 % [44]: 10.48 pJ per weight per row (7.05–11.17 pJ over that range; 5.51–14.15 pJ over the 30–55 % range of our original plan).

**Projection.** With the repaired pulse counts (resets counted as applied, the pulse reset's extra reads included), the GPU-to-analog ratio per weight per row is 31.5–32.4× at the 12.4 TOPS/W chip figure and 17.9–18.1× at the 6.94 TOPS/W system figure, for every published device and

Table 13: Conditional energy ratios for cells that pass the stated non-inferiority margin under ideal reads: the GPU-to-analog energy ratio per weight per row (12.4 TOPS/W measured chip figure / 6.94 TOPS/W projected system figure of the same inference chip, which includes estimated digital processing), printed only where the cell PASSES against the baseline; fail = the cell FAILS the margin; n.e. = the criterion is not established (UNRESOLVED). No finite-read operating point is demonstrated.

| | | MNIST | | CIFAR | | LM | |
|---|---|---|---|---|---|---|---|
| Device | Alg. | FP-SGD | Adam | FP-SGD | Adam | FP-SGD | Adam |
| Li-ECRAM | SGD | fail | fail | fail | fail | fail | fail |
| Li-ECRAM | TTv2 | 32 / 18 | 32 / 18 | fail | fail | n.e. | fail |
| Li-ECRAM | c-TTv2 | 32 / 18 | 32 / 18 | 32 / 18 | 32 / 18 | 32 / 18 | fail |
| Li-ECRAM | AGAD | 32 / 18 | 32 / 18 | 32 / 18 | 32 / 18 | 32 / 18 | fail |
| MO-ECRAM | SGD | fail | fail | fail | fail | fail | fail |
| MO-ECRAM | TTv2 | n.e. | n.e. | fail | fail | fail | fail |
| MO-ECRAM | c-TTv2 | 32 / 18 | 32 / 18 | n.e. | n.e. | fail | fail |
| MO-ECRAM | AGAD | 32 / 18 | 32 / 18 | 32 / 18 | 32 / 18 | fail | fail |
| Protonic-inspired | SGD | fail | fail | fail | fail | fail | fail |
| Protonic-inspired | TTv2 | 32 / 18 | 32 / 18 | fail | fail | n.e. | fail |
| Protonic-inspired | c-TTv2 | 32 / 18 | 32 / 18 | 32 / 18 | n.e. | 32 / 18 | fail |
| Protonic-inspired | AGAD | 32 / 18 | 32 / 18 | 32 / 18 | 32 / 18 | 32 / 18 | fail |
| W/$WO_x$ ECRAM | SGD | fail | fail | fail | fail | fail | fail |
| W/$WO_x$ ECRAM | TTv2 | 32 / 18 | 32 / 18 | fail | fail | fail | fail |
| W/$WO_x$ ECRAM | c-TTv2 | 32 / 18 | 32 / 18 | fail | fail | fail | fail |
| W/$WO_x$ ECRAM | AGAD | 32 / 18 | 32 / 18 | fail | fail | fail | fail |

algorithm. The reads are at least 97.0 % of the analog total: the ratio is set almost entirely by the two borrowed efficiencies, and changes in the simulated pulse counts barely move it.

**Conditional ratios for cells that pass the margin.** A low energy per operation is worth nothing without the accuracy. PASS means that the whole confidence interval of the gap lies within the stated non-inferiority margin under ideal reads, not that the losses are equal; we therefore print the ratio only for cells that PASS against the stated baseline (Table 13), mark FAIL cells as missing the margin and UNRESOLVED cells as "criterion not established":

- against FP-SGD: 11, 5 and 4 of the 16 published-device cells per task (MNIST, CIFAR, LM);
- against Adam: 11, 4 and 0 cells. **On the LM no analog cell passes the margin against Adam, so no conditional ratio exists against the stronger digital baseline.**

The comparison is at a fixed training budget; reaching a quality at a different number of steps is not modelled. No joint finite-read operating point of accuracy and energy, and no energy-to-target training trajectory, is demonstrated; finite precision, tiling and utilisation, reference maintenance, the digital TT state, update scheduling and full-system costs stay outside the ratio.

**What 100× would need.** At the central GPU figure, a 100× ratio needs reads at about 38.6–42.2 TOPS/W including their periphery, several times the chip figure above.

**The memistor.** Widrow's macroscopic memistor moves tens of microjoules of plating charge per level; its ratio is at most $8.0 \times 10^{-4}$ (far worse than the GPU), as expected for wet, centimetre-scale cells.

# 6 Discussion

**What the repaired simulations say, within their limits.**

- **Against FP-SGD, two device specifications reach the line on all three tasks** with a chopped Tiki-Taka algorithm: the AIHWKIT Li-ECRAM fit and the protonic-inspired scenario, each with c-TTv2 or AGAD (4 pairings).
- **Against Adam, none does.** On the language model the best analog cell is 23.36 % worse in bits per byte; the analog results inherit FP-SGD's optimiser gap (25.35 % on this model). Closing it would need analog-compatible adaptive updates, which this study does not test.
- **Plain SGD at the FP-SGD learning rate** exceeded the margin on every published device and task we ran (12 of 12 cells). This is a statement about that protocol, not about SGD after fair tuning.
- **The algorithm settings use the device.** On an ideal symmetric 1000-level device the selected TT settings lose far more than on the saturating devices, so in this configuration the accumulator's decay comes from the device's saturation.
- **Width** is not resolved: after per-arm tuning no arm's gap grows detectably beyond the margin over a factor of four, endpoint equivalence was established for one CIFAR arm only, a paired-by-seed analysis detects growth for another without resolving its size against the margin (growth beyond it is not excluded there or for Li-ECRAM TTv2), and the LM's shrinking gap reflects a weaker FP-SGD baseline at the largest width.
- **Energy** is a conditional projection set almost entirely by borrowed read efficiencies, and it is reported only for cells that pass the stated margin under ideal reads; against Adam on the LM no cell does.

**Widrow's memistor.** Taken at its inventor's description (thousands of near-linear levels), an optimistic historical-device scenario attained useful accuracy in this simulator; the pessimistic twin does not. Because its noise, asymmetry, drift and variability are largely unknown, this does not settle the accuracy of the real 1960 device. Its energy per level, tens of microjoules, is many orders of magnitude worse than today's devices.

## 6.1 Limitations

- **Simulation only.** Every device is a parametric model; the protonic entry is a scenario with estimated noise, asymmetry and saturation, and its finite retention is replaced by no drift. A 2026 measurement on BEOL-compatible protonic ECRAM [8] reports 3 % cycle-to-cycle and 21 % device-to-device variability; we did not refit the catalog to it, and we could not check those values against the full text. Estimated parameters are single points, not propagated ranges.
- **Ideal reads.** Read noise, IR drop, ADC/DAC quantisation and output bounds are not modelled, while the energy projection prices finite-precision reads.
- **The update surrogate.** Pulses of one batch are aggregated and solved as uniformly interleaved; discrete pulses do not commute, per-cell Poisson draws ignore row/column correlations, and the pulse budget clips the TT family's $A$ updates by up to 57.5 % in some

layers. Against explicit pulse streams (Section 4.3) the surrogate meets the registered lines on 10 of 12 update probes (point estimates), fails on the most clipped layer and does not reproduce the dependence of cells that share a line; in short training comparisons on MNIST and CIFAR's fully connected layers one label loses PASS, two closely spaced cells swap order and one failing SGD cell moves by more than 1 pp. The convolutional layers and the LM's training are not compared, and the comparison stream is itself a per-sample approximation of the literal one.

- **The reset.** The pulse reset uses a read and at most 62 pulses per weight; it leaves a residual (mean over runs 67.00 nominal steps, pooled over all reset events 26.97) and its latency and endurance cost are not modelled.
- **Algorithms.** TTv1, TTv2, c-TTv2 and AGAD were implemented from the papers' text; our TTv2 is a local variant (multi-pulse transfer, buffer remainder). The settings were selected once, on MNIST, with a mid device.
- **Small models and short budgets.** $10^5$ to $10^6$ weights, a 10,000-image CIFAR subset, a two-layer byte LM, a factor of four in width, fixed training budgets.
- **Statistics.** Five seeds (three in the sensitivity sheet and the diagnostic), no multiplicity correction, and best cells chosen among many; several cells remain UNRESOLVED.
- **Plans.** The plans were committed to our repository before the runs; they are not registered in an external registry.

# 7 A proposed roadmap for groups with device and fabrication resources

The simulations suggest what to measure first. None of the criteria below is an established device qualification: each is conditional on this simulator (ideal reads, the pulse surrogate, small models) and on the chopped Tiki-Taka setting that passed in Section 4.2.

**Stage 1: single-device characterisation** (tens of devices, no array).

- **Measurement.** At least 1000 up and 1000 down pulses at the intended width and voltage, repeated at least three times on at least 32 devices, with the read current measured (not inferred) and incomplete pulses rejected. Fit: step against state (saturation $\gamma_{\pm}$), up/down asymmetry, cycle-to-cycle noise (separated from read noise), device-to-device spreads, the symmetry point and the signed weight range around it, and drift.
- **What the simulations support** (c-TTv2 sheet, CIFAR $c = 32$, absolute 2 pp line, three seeds, Table 12): a step asymmetry up to 0.2 and a reference error up to 0.05 of the half range, one axis at a time around the protonic-inspired base. Thresholds for levels, saturation, write noise and retention are not established by this grid; what it shows is where training clearly fails (no saturation or $\gamma \geq 1$, reference error 0.2, retention of $10^3$ updates or less). A device group should therefore measure all of these and simulate its own fitted device, rather than read pass lines off this sheet. Combined deviations are not covered.
- **Endurance and pulse budget.** Report per-array and per-layer pulse distributions and the hottest cells, not only averages; the TT family's $A$ arrays and the resets dominate the write load, and the pulse budget per update clips the largest layers (Section 4.2).

**Stage 2: the symmetry point and the reset as physical processes.**

- **Measurement.** The pulse count, latency and energy to bring a device to its symmetry point or to its read zero; the residual error, its spread and its drift over hours and $10^6$ cycles; the signed range left on each side.
- **Why.** With a realistic pulse budget the reset is incomplete (on the protonic-inspired device, 62 alternating pulses leave 97.55 % of a displacement), so the reset is a device specification, not a free operation.

**Stage 3: small arrays with hardware in the loop.**

- **Hardware.** Two $64 \times 64$ arrays per layer ($A$ and $C$); forward and backward passes on the host from the read weights; updates on chip as stochastic pulse coincidences; resets or chopping as the algorithm requires.
- **Prediction first.** Before training, fit the Stage 1 model on several independent chips, run this simulator with the fit, and state a *prediction interval* that includes device-population variation, fit uncertainty, training randomness and measurement noise (not a confidence interval of the simulated mean, which shrinks with more seeds). The hardware should land inside it; independently, within 2 pp of FP-SGD shows the device works.

**Stage 4: full arrays and in-memory reads.** Move the forward and backward reads on chip and measure the system energy per weight per row, with the read precision that gives the accuracy, against the conditional projection of Section 5; convert retention requirements into time with the full step duration.

**The 1960 memistor.** Appendix A gives a proposed, unbuilt bench replication protocol with hobby parts and safety notes. Before its fits are used as metrology it must measure the actual read current, keep per-pulse charge, separate read from write noise and reject incomplete pulses.

# 8 Conclusion

In this simulator, with ideal reads and every update made of device pulses, two electrochemical specifications (the AIHWKIT Li-ECRAM fit and a protonic-inspired scenario) train three small networks to within the pre-set margins of SGD with exact updates when a chopped Tiki-Taka algorithm moves the gradient; plain SGD at the same learning rate does not on any published device. Against a tuned digital Adam none of them does, and on the language model the best is 23.36 % worse in bits per byte. After per-arm tuning no arm's gap grows detectably beyond the margin over a factor of four in width; endpoint equivalence was established for one CIFAR pairing only, and a paired-by-seed analysis detects growth for another, of a size not resolved against the margin. The energy advantage is a conditional projection (17.9–32.4× per weight-row at borrowed inference-chip read efficiencies), reported only for cells that pass the margin under ideal reads. After an independent review we repaired the simulator and report how much each repair moved the original results. The remaining questions are computational and methodological (the fidelity of the update surrogate beyond the cases tested here, dependence-aware statistics, the construction of controls) as well as experimental (the reset, the reference and the read path in real arrays).

## 9 Code and data availability

Nothing has been deposited publicly yet. On deposit, one archive will hold everything needed to reproduce this paper, with versions, source hashes, dataset digests and split offsets, run manifests and licences:

- the plans (`PLAN-PAPER.md`, `PLAN-REVISION.md`, `PLAN-REVISION-2.md`, `PLAN-REVISION-3.md`, `PLAN-REVISION-4.md`), their deviations (`DEVIATIONS-PAPER.md`) and the responses to the four independent reviews (`RESPONSE-TO-REVIEW.md`, `RESPONSE-TO-REVIEW-2.md`, `RESPONSE-TO-REVIEW-3.md`, `RESPONSE-TO-REVIEW-4.md`);
- the original simulator and the repaired one (`sim/*_rev.py`), the explicit pulse streams and probes (`sim/*_rev2.py`, `sim/*_rev3.py`, `sim/sweep_rev4.py`), with their checks;
- the device catalog with per-field provenance;
- every run record (one JSON file per training run, with its configuration, metric, pulse counts and hardware);
- the scripts that derive every number, table and figure of this paper from those records (`analyze.py`, `make_figures.py`, `energy/energy_system.py`; for the revision `revision/analyze_rev.py`, `revision/analyze_d1.py`, `revision/compare_curves.py`, `energy/energy_rev.py`, `revision2/analyze_rev2.py`, `revision3/analyze_rev3.py`, `revision4/analyze_rev4.py`);
- the saved pulse draws of the update probes: 12 NumPy archives (81,905,772 bytes, outside the code repository; each file's SHA-256 is in its probe record and in `PUBLISHING.md`), from which `revision4/rebuild_draw_tables.py` recomputes every statistic of Tables 6–8 and checks the rebuilt tables against the paper's byte for byte;
- the citation check (every reference verified against Crossref, the arXiv API or the live URL; `records/CITE-CHECK.json`);
- the bench protocol.

The archive DOI will be added on deposit (Zenodo). Data: MNIST, CIFAR-10 and enwik8 are public and pinned by digest. All text data used for training is human-written (Wikipedia, 2006).

## Author contributions and use of AI

Questeria conceived and directed the study. Claude (Anthropic), an AI system, wrote the simulator and its checks, designed and pre-registered the experiments under Questeria's direction, ran them on rented cloud GPUs, analysed the records, verified the references, and drafted the text.

Under the policies of arXiv [45] and of most journals, an AI system cannot be an author or take responsibility for a paper. For any formal submission the arXiv-compliant variant of this paper lists Questeria as the sole author and states Claude's role in this section. All numbers were produced by scripts from the saved records.

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## References

[1] Bernard Widrow. An adaptive "ADALINE" neuron using chemical "memistors". Technical Report 1553-2, Stanford Electronics Laboratories, Stanford University, 1960. Scanned copy at https://isl.stanford.edu/ widrow/papers/t1960anadaptive.pdf (image only; title not machine-checked).

[2] B. WIDROW and M. E. HOFF. Adaptive switching circuits. Technical report, Defense Technical Information Center, 1960.

[3] B. Widrow and M.A. Lehr. 30 years of adaptive neural networks: perceptron, madaline, and backpropagation. *Proceedings of the IEEE*, 78(9):1415–1442, 1990. doi: 10.1109/5.58323.

[4] Jianshi Tang et al. Ecram as scalable synaptic cell for high-speed, low-power neuromorphic computing. In *2018 IEEE International Electron Devices Meeting (IEDM)*, pages 13.1.1–13.1.4, 2018. doi: 10.1109/iedm.2018.8614551.

[5] Elliot J. Fuller, Farid El Gabaly, François Léonard, Sapan Agarwal, Steven J. Plimpton, Robin B. JacobsGedrim, Conrad D. James, Matthew J. Marinella, and A. Alec Talin. Li-ion synaptic transistor for low power analog computing. *Advanced Materials*, 29(4):1604310, 2016. doi: 10.1002/adma.201604310.

[6] Seyoung Kim et al. Metal-oxide based, cmos-compatible ecram for deep learning accelerator. In *2019 IEEE International Electron Devices Meeting (IEDM)*, pages 35.7.1–35.7.4, 2019. doi: 10.1109/iedm19573.2019.8993463.

[7] Murat Onen, Nicolas Emond, Baoming Wang, Difei Zhang, Frances M. Ross, Ju Li, Bilge Yildiz, and Jesús A. del Alamo. Nanosecond protonic programmable resistors for analog deep learning. *Science*, 377(6605):539–543, 2022. doi: 10.1126/science.abp8064.

[8] Longlong Xu, Mantao Huang, Stephen D. Funni, Judy J. Cha, and Bilge Yildiz. Quantifying variability, symmetry, and nonlinearity of protonic electrochemical random-access memory for hardware neural network training. *ACS Nano*, 20(39):26566–26579, 2026. doi: 10.1021/acsnano.6c09861.

[9] Peng Chen, Fenghao Liu, Peng Lin, Peihong Li, Yu Xiao, Bihua Zhang, and Gang Pan. Open-loop analog programmable electrochemical memory array. *Nature Communications*, 14(1): 6184, 2023. doi: 10.1038/s41467-023-41958-4.

[10] Tayfun Gokmen and Yurii Vlasov. Acceleration of deep neural network training with resistive cross-point devices: Design considerations. *Frontiers in Neuroscience*, 10, 2016. doi: 10.3389/fnins.2016.00333.

[11] Tayfun Gokmen and Wilfried Haensch. Algorithm for training neural networks on resistive device arrays. *Frontiers in Neuroscience*, 14:103, 2020. doi: 10.3389/fnins.2020.00103.

[12] Tayfun Gokmen. Enabling training of neural networks on noisy hardware. *Frontiers in Artificial Intelligence*, 4:699148, 2021. doi: 10.3389/frai.2021.699148.

[13] Malte J. Rasch, Fabio Carta, Omobayode Fagbohungbe, and Tayfun Gokmen. Fast and robust analog in-memory deep neural network training. *Nature Communications*, 15(1):7133, 2024. doi: 10.1038/s41467-024-51221-z.

[14] Zhaoxian Wu, Quan Xiao, Tayfun Gokmen, Omobayode Fagbohungbe, and Tianyi Chen. Analog in-memory training on general non-ideal resistive elements: The impact of response functions. *arXiv preprint arXiv:2502.06309*, 2025.

[15] Zhaoxian Wu, Tayfun Gokmen, Omobayode Fagbohungbe, T. Patrick Xiao, and Tianyi Chen. Making analog training scale: Co-designing mapping, optimizer, and converters. *arXiv preprint arXiv:2609.36584*, 2026.

[16] Blaise Mouttet. Memristive systems analysis of 3-terminal devices. *arXiv preprint arXiv:1012.5124*, 2010.

[17] L. Chua. Memristor-the missing circuit element. *IEEE Transactions on Circuit Theory*, 18(5): 507–519, 1971. doi: 10.1109/tct.1971.1083337.

[18] Dmitri B. Strukov, Gregory S. Snider, Duncan R. Stewart, and R. Stanley Williams. The missing memristor found. *Nature*, 453(7191):80–83, 2008. doi: 10.1038/nature06932.

[19] B. R. Gaines. Stochastic computing. In *Proceedings of the April 18-20, 1967, spring joint computer conference on - AFIPS '67 (Spring)*, page 149, 1967. doi: 10.1145/1465482.1465505.

[20] Sapan Agarwal, Steven J. Plimpton, David R. Hughart, Alexander H. Hsia, Isaac Richter, Jonathan A Cox, Conrad D. James, and Matthew J. Marinella. Resistive memory device requirements for a neural algorithm accelerator. In *2016 International Joint Conference on Neural Networks (IJCNN)*, pages 929–938, 2016. doi: 10.1109/ijcnn.2016.7727298.

[21] Tayfun Gokmen, Murat Onen, and Wilfried Haensch. Training deep convolutional neural networks with resistive cross-point devices. *Frontiers in Neuroscience*, 11:538, 2017. doi: 10.3389/fnins.2017.00538.

[22] Geoffrey W. Burr et al. Neuromorphic computing using non-volatile memory. *Advances in Physics: X*, 2(1):89–124, 2016. doi: 10.1080/23746149.2016.1259585.

[23] Daniele Ielmini and H.-S. Philip Wong. In-memory computing with resistive switching devices. *Nature Electronics*, 1(6):333–343, 2018. doi: 10.1038/s41928-018-0092-2.

[24] Abu Sebastian, Manuel Le Gallo, Riduan Khaddam-Aljameh, and Evangelos Eleftheriou. Memory devices and applications for in-memory computing. *Nature Nanotechnology*, 15(7): 529–544, 2020. doi: 10.1038/s41565-020-0655-z.

[25] Wilfried Haensch, Tayfun Gokmen, and Ruchir Puri. The next generation of deep learning hardware: Analog computing. *Proceedings of the IEEE*, 107(1):108–122, 2019. doi: 10.1109/jproc.2018.2871057.

[26] S. R. Nandakumar et al. Mixed-precision deep learning based on computational memory. *Frontiers in Neuroscience*, 14:406, 2020. doi: 10.3389/fnins.2020.00406.

[27] Malte J. Rasch, Fabio Carta, Omebayode Fagbohungbe, and Tayfun Gokmen. Fast offset corrected in-memory training. *arXiv preprint arXiv:2303.04721*, 2023.

[28] Malte J. Rasch, Diego Moreda, Tayfun Gokmen, Manuel Le Gallo, Fabio Carta, Cindy Goldberg, Kaoutar El Maghraoui, Abu Sebastian, and Vijay Narayanan. A flexible and fast pytorch toolkit for simulating training and inference on analog crossbar arrays. In *2021 IEEE 3rd International Conference on Artificial Intelligence Circuits and Systems (AICAS)*, pages 1–4, 2021. doi: 10.1109/aicas51828.2021.9458494.

[29] Elliot J. Fuller et al. Parallel programming of an ionic floating-gate memory array for scalable neuromorphic computing. *Science*, 364(6440):570–574, 2019. doi: 10.1126/science.aaw5581.

[30] Yoeri van de Burgt, Ewout Lubberman, Elliot J. Fuller, Scott T. Keene, Grégorio C. Faria, Sapan Agarwal, Matthew J. Marinella, A. Alec Talin, and Alberto Salleo. A non-volatile organic electrochemical device as a low-voltage artificial synapse for neuromorphic computing. *Nature Materials*, 16(4):414–418, 2017. doi: 10.1038/nmat4856.

[31] Yiyang Li, T. Patrick Xiao, Christopher H. Bennett, Erik Isele, Armantas Melianas, Hanbo Tao, Matthew J. Marinella, Alberto Salleo, Elliot J. Fuller, and A. Alec Talin. In situ parallel training of analog neural network using electrochemical random-access memory. *Frontiers in Neuroscience*, 15:636127, 2021. doi: 10.3389/fnins.2021.636127.

[32] M. Prezioso, F. Merrikh-Bayat, B. D. Hoskins, G. C. Adam, K. K. Likharev, and D. B. Strukov. Training and operation of an integrated neuromorphic network based on metal-oxide memristors. *Nature*, 521(7550):61–64, 2015. doi: 10.1038/nature14441.

[33] Stefano Ambrogio et al. Equivalent-accuracy accelerated neural-network training using analogue memory. *Nature*, 558(7708):60–67, 2018. doi: 10.1038/s41586-018-0180-5.

[34] Peng Yao, Huaqiang Wu, Bin Gao, Jianshi Tang, Qingtian Zhang, Wenqiang Zhang, J. Joshua Yang, and He Qian. Fully hardware-implemented memristor convolutional neural network. *Nature*, 577(7792):641–646, 2020. doi: 10.1038/s41586-020-1942-4.

[35] Weier Wan et al. A compute-in-memory chip based on resistive random-access memory. *Nature*, 608(7923):504–512, 2022. doi: 10.1038/s41586-022-04992-8.

[36] S. Ambrogio et al. An analog-ai chip for energy-efficient speech recognition and transcription. *Nature*, 620(7975):768–775, 2023. doi: 10.1038/s41586-023-06337-5.

[37] Manuel Le Gallo et al. A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference. *Nature Electronics*, 6(9):680–693, 2023. doi: 10.1038/s41928-023-01010-1.

[38] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998. doi: 10.1109/5.726791.

[39] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[40] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. *arXiv preprint arXiv:1706.03762*, 2017.

[41] Matt Mahoney. About the test data (enwik8 and enwik9), large text compression benchmark. `http://mattmahoney.net/dc/textdata.html`, 2011. accessed 2026-10-25.

[42] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

[43] NVIDIA Corporation. NVIDIA H100 Tensor Core GPU datasheet. `https://www.nvidia.com/en-us/data-center/h100/`, 2023. accessed 2026-10-25.

[44] Aaron Grattafiori et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[45] arXiv. arxiv moderation policies and procedures (including the policy on authors' use of generative ai tools). `https://info.arxiv.org/help/moderation/index.html`, 2026. accessed 2026-10-25.

# A Bench replication of the 1960 memistor

This appendix is a protocol for others to run. We built nothing and bought nothing. It is meant to measure, with hobby parts, the properties that Widrow's reports leave open (Section 2): levels, up/down symmetry, linearity, write noise, drift and energy per step. The measured cell can then be placed on the same map as the modern devices. The full protocol, an Arduino sketch, a parts list with estimated prices, and an analysis script that turns the log into a device specification for our simulator are in the repository (`BENCH.md`, `bench/`).

**Pass criteria.** The cell must show at least 32 near-linear, symmetric levels on one lead: over at least 32 consecutive equal-charge pulses in each direction,

1. the mean up step and the mean down step agree within 10%;
2. the step varies by less than $\pm 30\%$ across the range ($\gamma \leq 0.3$);
3. the pulse-to-pulse scatter is below half a step ($c_{2c} \leq 0.5$).

A four-input ADALINE trained by LMS on four such leads should then cut its mean squared error at least tenfold.

**Build.**

- Electrolyte: 0.5 M copper(II) sulphate in distilled water. Sulphuric acid and brighteners are deliberately left out for safety, so expect some hysteresis.
- Electrodes: 2 mm HB graphite leads with wrapped-wire contacts, the lower contact insulated with nail polish, a wetted length of about 20 mm, and a bare copper counter electrode.
- Electronics: an Arduino Nano, an INA219 current/voltage sensor, and a CD4052 multiplexer for four-wire reads.

**Protocol.**

1. Read-noise baseline.
2. Plating range in 1 mC pulses.
3. 64 pulses down and 64 up from mid-range, repeated, to measure the step against conductance, the asymmetry and the scatter.
4. A two-hour (ideally overnight) drift run, covered and then stirred, to separate oxygen-driven corrosion.
5. The integral of cell voltage × current per pulse, which gives the energy per step.
6. 20 full cycles to measure endurance.
7. The LMS ADALINE.

**Safety.**

- Copper sulphate is harmful if swallowed, irritates eyes and skin, and is very toxic to aquatic life. Wear splash goggles and nitrile gloves, and work on a tray away from food, children and pets.
- Use low voltage only (USB). Keep electronics out of splash range, keep aluminium away from the solution, and reduce the current if bubbles form.
- Never pour waste down the drain. Take it to household hazardous-waste collection.
- For eye contact, rinse for 15 minutes and seek medical advice.

**What the bench cannot show.** A pencil-lead memistor is macroscopic. Its energy per step is set by the copper moved, about 15 $\mu$J per level by Widrow's figures and 986 $\mu$J for our pencil-lead estimate. That is many orders of magnitude above an H100's energy per weight update (Section 5). The bench can confirm or refute the linearity and symmetry the memistor promised; it cannot make training cheaper. The cheap direction, if any, is the nanoscale descendants.

# B Hyperparameter selection

Learning rates and algorithm settings were selected on validation splits only, with seed 0.

- **MNIST learning rate.** The FP grid was $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$, extended by the edge rule. The choice is 0.003.
- **TT-family settings.** They were chosen on MNIST validation with a mid-quality device (128 levels, asymmetry 0.1, $\gamma$ 0.3, $c_{2c}$ 0.5, $\sigma_r$ 0.02):
  - TTv2: $\lambda \in \{0.1, 0.3, 1, 3\}$ and a period of 16 or 64 batches, with or without a matched $A$ learning rate; in the revision also with the pulse reset or none;
  - c-TTv2: $\lambda \in \{0.1, 0.3\}$, $\rho \in \{0.05, 0.2\}$, with the pulse reset or none;
  - AGAD: $\lambda \in \{0.1, 0.3\}$, $\rho \in \{0.05, 0.2\}$, $\beta \in \{0.3, 1\}$.

Table 14: Revised settings (repaired simulator), chosen on MNIST validation with the mid device, seed 0; every E3, E4 and E5 record carries exactly these settings. All TT-family runs use BL 31 pulse slots per sample and at most 31 transfer pulses per weight. Better chopped algorithm: c-TTv2; S-ALGO: TTv2.

| Alg. | Setting | $A$ reset | Val. % |
|---|---|---|---|
| TTv2 | $\lambda$ 0.1, $A$ lr $\times$10, period 64 | pulse (at most 62 pulses, 2 read-correct iterations) | 97.6 |
| c-TTv2 | $\lambda$ 0.1, $A$ lr $\times$10, period 16, $\rho$ 0.05 | pulse (at most 62 pulses, 2 read-correct iterations) | 97.6 |
| AGAD | $\lambda$ 0.1, $A$ lr $\times$10, period 16, $\rho$ 0.2, $\beta$ 0.3 | none | 97.3 |

Learning rates. FP-SGD: MNIST 0.003; CIFAR $c = 16/32/64$: 0.0003/0.001/0.0003; LM $d = 64/128/256$: 0.0001/0.0001/3e-05. Adam: MNIST 0.001, CIFAR 0.001, LM 0.003. Width arms (E4), narrow/mid/wide: CIFAR Li-ECRAM c-TTv2: 0.0003/0.003/0.003; CIFAR Li-ECRAM TTv2: 0.0003/0.0003/0.0003; CIFAR Protonic-inspired c-TTv2: 0.003/0.003/0.003; CIFAR Protonic-inspired TTv2: 0.0003/0.0003/0.003; LM Li-ECRAM c-TTv2: 0.0001/0.0003/0.0001; LM Li-ECRAM TTv2: 0.0001/0.0003/0.0001; LM Protonic-inspired c-TTv2: 0.0003/0.0003/0.0001; LM Protonic-inspired TTv2: 0.0003/0.0003/0.0001.

- **Derived choices.** The better chopped algorithm (c-TTv2) is used in the scale study, and the best of the three (TTv2) in the specification sheet.
- **CIFAR and LM learning rates** were chosen per width (Section 4.4); Adam's on a grid of $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ per task with the edge rule.

Table 14 gives the revised selections that every revised run uses (the original selection table is Table 18, Appendix E).

# C Simulator checks

Each check runs twice. On the real code it must pass. Once for every planted fault it must fail, which shows that the check catches that fault. A crash under a fault also counts as caught. Both suites ran on the desk before the plan was committed, and again at the start of every cloud session; any failure would have stopped the session.

Table 15: Simulator checks and the planted faults each one catches (records/TESTS-PAPER.json, records/TESTS-laneME-checks-on-paper-sim.json).

| Check | Result | Planted faults caught |
|---|---|---|
| `c_asym_drift` | pass | asymmetry ignored |
| `c_c2c_noise` | pass | write noise $\sim$ n not sqrt(n) |
| `c_constant_step_reference` | pass | constant-step reference at a bound (the D1 bug) |
| `c_drift` | pass | drift relaxes to 0 not bare |
| `c_fp_equivalence` | pass | apply_exact sign flipped |
| `c_level_step` | pass | step size halved |
| `c_pulse_mean` | pass | pulse slots BL dropped; pulse up/down swapped |
| `c_symmetry_point` | pass | symmetry point sign |
| `c_ttv2_learns` | pass | TT transfer sign flipped |
| `c_family_learns` | pass | A-update inputs not chopped |
| `c_gamma_d2d_sp` | pass | gamma spread shared by the whole tile |
| `c_identity_sgd_ttv2` | pass | down-branch saturation uses gamma_up |
| `c_offset_rejection` | pass | AGAD reference never set; chopper demodulation missing; chopper never flips |
| `c_ttv1_direction` | pass | TTv1 transfer sign flipped |

# D Earlier development results

This study began as an exploratory screen with the same simulator (SGD and TTv2 only, 1–3 seeds). The screen is summarised here because some of its observations motivated the pre-registered experiments:

- a CIFAR gap "just over 2 pp";
- a gap that "grows with width";
- a "pencil lead built today" that fails every task.

The earlier runs differ from this paper's in three ways:

- MNIST ran on a different GPU;
- the LM's selection split overlapped its training data (corrected here, Section 3.5);
- CIFAR and the LM had one or two seeds.

With five seeds the earlier point estimates are mostly confirmed (Table 16), but the earlier pass/fail readings were not meaningful at that sample size. The CIFAR baseline's seed-to-seed spread is comparable to the gap line itself.

The pessimistic memistor and the pencil-lead estimate were run only in the earlier screen. Both fail every task with SGD and with TTv2. Their specifications are estimates; the bench protocol of Appendix A would replace the pencil-lead estimate by measurements.

Table 16: Earlier development results (lane ME: same simulator, 1–3 seeds, MNIST on a different GPU, the LM selected on an overlapping split) beside this paper's pre-registered 5-seed values. Gaps in pp (MNIST, CIFAR) and % (LM). The pessimistic memistor and the pencil lead appear only in the earlier runs.

| | | MNIST | | CIFAR | | LM | |
|---|---|---|---|---|---|---|---|
| Device | Alg. | earlier | this paper | earlier | this paper | earlier | this paper |
| Li-ECRAM (Tang 2018) | SGD | 5.24 (n=3) | 4.85 (n=5) | 21.78 (n=1) | 20.81 (n=5) | 39.75 (n=1) | 45.58 (n=5) |
| Li-ECRAM (Tang 2018) | TTv2 | 1.17 (n=3) | 1.32 (n=5) | 2.15 (n=2) | 2.95 (n=5) | 4.08 (n=2) | 3.52 (n=5) |
| Protonic (Onen 2022) | SGD | 3.67 (n=3) | 3.07 (n=5) | 19.54 (n=1) | 15.77 (n=5) | 37.62 (n=1) | 40.82 (n=5) |
| Protonic (Onen 2022) | TTv2 | 0.77 (n=3) | 1.04 (n=5) | 2.13 (n=2) | 2.13 (n=5) | 4.21 (n=2) | 3.64 (n=5) |
| MO-ECRAM (Kim 2019) | SGD | 7.20 (n=3) | 6.96 (n=5) | 26.19 (n=1) | 25.29 (n=5) | 76.49 (n=1) | 50.84 (n=5) |
| MO-ECRAM (Kim 2019) | TTv2 | 1.84 (n=3) | 1.80 (n=5) | 8.04 (n=1) | 5.45 (n=5) | 17.19 (n=1) | 16.51 (n=5) |
| W/YSZ/$WO_x$ (Chen 2023) | SGD | 8.19 (n=3) | 8.25 (n=5) | 37.88 (n=1) | 36.82 (n=5) | 54.79 (n=1) | 50.67 (n=5) |
| W/YSZ/$WO_x$ (Chen 2023) | TTv2 | 1.49 (n=3) | 1.48 (n=5) | 10.45 (n=1) | 10.59 (n=5) | 20.77 (n=1) | 21.17 (n=5) |
| Memistor (Widrow 1960, as claimed) | SGD | 1.71 (n=3) | 1.29 (n=5) | 17.15 (n=1) | 9.86 (n=5) | 28.01 (n=1) | 30.94 (n=5) |
| Memistor (Widrow 1960, as claimed) | TTv2 | 0.63 (n=3) | 0.73 (n=5) | 3.24 (n=1) | 2.86 (n=5) | 10.75 (n=1) | 8.99 (n=5) |
| Memistor (pessimistic) | SGD | 32.58 (n=3) | – | 41.52 (n=1) | – | 47.26 (n=1) | – |
| Memistor (pessimistic) | TTv2 | 15.33 (n=3) | – | 39.16 (n=1) | – | 27.44 (n=1) | – |
| Pencil lead built today | SGD | 23.05 (n=3) | – | 43.92 (n=1) | – | 42.36 (n=1) | – |
| Pencil lead built today | TTv2 | 11.72 (n=3) | – | 37.39 (n=1) | – | 21.98 (n=1) | – |

# E The original results (superseded)

The first version of this paper reported the experiments below, run on the original simulator. An independent review found that this simulator (i) drew the initialisation, the data order and all device randomness from one shared random stream, so that two arms of the same seed did not share their initial weights or their data order; (ii) moved the TTv2 and c-TTv2 accumulator straight to its symmetry point while charging 62 pulses it did not apply; and (iii) copied the AIHWKIT preset fields into a model with different bounds and no slope spread. The results are kept here unchanged, as preliminary results of that simulator; the revised experiments of Sections 4.2–4.5 replace them. The diagnostic of Appendix F measures how much each repair moves them.

The original headline readings were: plain SGD failed on every published device and task; the Li-ECRAM fit and the protonic device passed or were unresolved on all three tasks with a Tiki-Taka-family algorithm; the original width classifier labelled every arm FLAT (which, as the review noted, is not evidence of equivalence); and the original specification sheet was mostly "not reachable", which led to a post hoc degradation sheet relative to an already imperfect base (a CIFAR base gap of 2.39 pp plus 2 pp admits gaps up to 4.39 pp). The best original analog LM cell was 21.66 % worse in bits per byte than digital Adam (a perplexity ratio of 1.425).

Table 17: Version 1 (superseded). Device catalog (simulator units; Section 3.1). Levels $n_{\text{states}}$, global asymmetry $a$, saturation $\gamma_+/\gamma_-$, write noise $c_{2c}$ (per pulse, relative to the step), step spread $\sigma_{\text{d2d}}$, drift time constant $\tau$ (training steps), reference error $\sigma_r$. Provenance per field is in records/CATALOG.json.

| Device | Label | $n_{\text{st}}$ | $a$ | $\gamma_+/\gamma_-$ | $c_{2c}$ | $\sigma_{\text{d2d}}$ | $\tau$ | $\sigma_r$ |
|---|---|---|---|---|---|---|---|---|
| Li-ECRAM (Tang 2018) | PUBLISHED-FIT | 1000 | 0 | 0.12/0.51 | 0.3 | 0.1 | none | 0.02 |
| MO-ECRAM (Kim 2019) | PUBLISHED-FIT | 7089 | 0 | 0.42/0.73 | 2 | 0.1 | none | 0.02 |
| Protonic (Onen 2022) | PARTLY-PUBLISHED | 1000 | 0.05 | 0.20/0.20 | 0.3 | 0.1 | none | 0.02 |
| W/YSZ/$WO_x$ (Chen 2023) | PUBLISHED-FIT | 256 | 0 | 0.63/0.63 | 2.048 | 0.023 | none | 0.02 |
| Memistor (Widrow 1960, as claimed) | CLAIM+ESTIMATE | 1000 | 0.05 | 0.10/0.10 | 0.3 | 0.2 | none | 0.02 |
| Memistor (pessimistic) | ESTIMATE | 100 | 0.2 | 0.30/0.30 | 1 | 0.3 | 1000 | 0.05 |
| Pencil lead built today | ESTIMATE | 100 | 0.1 | 0.30/0.30 | 0.5 | 0.2 | 940 | 0.05 |

Table 18: Version 1 (superseded). Settings chosen on MNIST validation with the mid device (Appendix B); validation accuracy in %.

| Algorithm | Chosen setting | Val. acc. |
|---|---|---|
| TTv2 | $\lambda$ 0.1, $A$ lr $\times$10, period 16 | 97.86 |
| c-TTv2 | $\lambda$ 0.1, $A$ lr $\times$10, period 16, $\rho$ 0.2, reset yes | 97.82 |
| AGAD | $\lambda$ 0.1, $A$ lr $\times$10, period 16, $\rho$ 0.05, $\beta$ 0.3, reset no | 97.18 |

Table 19: Version 1 (superseded). Main result (E3): gap to the exact-update baseline, mean and 95% t-interval over 5 seeds, paired by seed. MNIST and CIFAR in percentage points of test accuracy; LM in % of bits per byte. P = the whole interval is below the line (2 pp; 5%), F = wholly above, U = the interval contains the line. Every arm uses the FP validation learning rate.

| Device | Algorithm | MNIST gap (pp) | CIFAR-10 subset gap (pp) | char-LM gap (%) |
|---|---|---|---|---|
| FP baseline (acc. %; bpb) | | 97.85 [97.79, 97.92] | 60.39 [58.64, 62.14] | 2.912 [2.822, 3.001] |
| Li-ECRAM (Tang 2018) | SGD | 4.85 [4.73, 4.98] F | 20.81 [18.06, 23.57] F | 45.58 [31.52, 59.64] F |
| | TTv2 | 1.32 [0.49, 2.14] U | 2.95 [0.94, 4.96] U | 3.52 [0.86, 6.18] U |
| | c-TTv2 | 1.79 [0.94, 2.65] U | 0.10 [-1.27, 1.47] P | -1.29 [-4.46, 1.87] P |
| | AGAD | 0.86 [0.67, 1.06] P | 1.06 [-1.05, 3.17] U | 3.60 [-0.09, 7.30] U |
| MO-ECRAM (Kim 2019) | SGD | 6.96 [6.60, 7.31] F | 25.29 [22.15, 28.44] F | 50.84 [36.24, 65.45] F |
| | TTv2 | 1.80 [1.68, 1.91] P | 5.45 [2.15, 8.75] F | 16.51 [13.17, 19.84] F |
| | c-TTv2 | 1.90 [1.81, 1.99] P | 1.94 [-0.15, 4.03] U | 15.68 [7.21, 24.14] F |
| | AGAD | 0.49 [0.33, 0.66] P | 1.39 [-0.31, 3.09] U | 8.88 [6.10, 11.65] F |
| Protonic (Onen 2022) | SGD | 3.07 [2.85, 3.29] F | 15.77 [10.75, 20.79] F | 40.82 [34.31, 47.33] F |
| | TTv2 | 1.04 [0.40, 1.69] P | 2.13 [0.49, 3.77] U | 3.64 [0.98, 6.30] U |
| | c-TTv2 | 0.74 [0.44, 1.05] P | 0.08 [-2.57, 2.73] U | -0.84 [-4.30, 2.62] P |
| | AGAD | 1.17 [0.73, 1.62] P | 0.56 [-0.85, 1.98] P | 3.82 [0.32, 7.33] U |
| W/YSZ/$WO_x$ (Chen 2023) | SGD | 8.25 [7.47, 9.02] F | 36.82 [33.17, 40.47] F | 50.67 [47.53, 53.81] F |
| | TTv2 | 1.48 [1.37, 1.60] P | 10.59 [8.94, 12.24] F | 21.17 [17.11, 25.24] F |
| | c-TTv2 | 1.75 [1.33, 2.17] U | 9.87 [7.90, 11.84] F | 21.23 [17.81, 24.66] F |
| | AGAD | 1.08 [0.79, 1.37] P | 17.55 [15.52, 19.58] F | 26.59 [20.81, 32.38] F |
| Memistor (Widrow 1960, as claimed) | SGD | 1.29 [1.12, 1.46] P | 9.86 [2.80, 16.92] F | 30.94 [25.31, 36.57] F |
| | TTv2 | 0.73 [0.28, 1.17] P | 2.86 [-0.64, 6.35] U | 8.99 [5.43, 12.54] F |
| | c-TTv2 | 0.49 [0.21, 0.77] P | 1.32 [0.08, 2.55] U | 6.59 [1.96, 11.23] U |
| | AGAD | 1.29 [0.64, 1.94] P | 2.06 [0.68, 3.44] U | 7.76 [3.53, 11.99] U |

Table 20: Version 1 (superseded). Scale (E4): every arm uses its own validation-chosen learning rate (in parentheses) at every width; gap (pp for CIFAR, % for the LM), mean and 95% interval over 3 seeds paired with FP at its own learning rate. Verdict by the pre-registered rule (Section 4.4).

| Task | Width | FP metric (lr) | Li-ECRAM c-TTv2 | Li-ECRAM TTv2 | Protonic c-TTv2 | Protonic TTv2 |
|---|---|---|---|---|---|---|
| CIFAR | 16 | 55.70 (0.0003) | 0.64 [-1.53, 2.82] (0.003) | 1.86 [-0.64, 4.35] (0.001) | -2.13 [-7.14, 2.88] (0.003) | 0.41 [-1.44, 2.26] (0.001) |
| CIFAR | 32 | 60.15 (0.001) | -0.40 [-3.13, 2.33] (0.001) | 2.12 [-1.66, 5.90] (0.001) | -1.88 [-6.35, 2.59] (0.003) | -0.21 [-5.27, 4.84] (0.003) |
| CIFAR | 64 | 60.87 (0.0003) | -1.34 [-5.57, 2.89] (0.001) | 1.11 [0.17, 2.05] (0.001) | -1.30 [-3.56, 0.96] (0.003) | 0.33 [-2.43, 3.09] (0.001) |
| | | verdict: | FLAT | FLAT | FLAT | FLAT |
| LM | 64 | 3.193 (0.0001) | -0.55 [-8.58, 7.48] (0.0001) | -0.31 [-6.26, 5.63] (0.0003) | -7.21 [-11.77, -2.65] (0.0003) | -3.45 [-6.44, -0.46] (0.0003) |
| LM | 128 | 2.881 (0.0001) | -0.09 [-7.23, 7.04] (0.0001) | -0.60 [-8.73, 7.53] (0.0003) | -6.38 [-12.53, -0.23] (0.0003) | -3.88 [-10.90, 3.14] (0.0003) |
| LM | 256 | 3.100 (3e-05) | -13.73 [-14.66, -12.79] (0.0001) | -9.01 [-10.21, -7.82] (0.0001) | -12.97 [-13.57, -12.36] (0.0001) | -8.48 [-9.67, -7.29] (0.0001) |
| | | verdict: | FLAT | SHRINKS | FLAT | SHRINKS |

Table 21: Version 1 (superseded). Device specification sheet (E5) for TTv2: the weakest grid value of each parameter that keeps the gap under the line (2 pp CIFAR; 5% LM) with every other parameter at the protonic base (levels 1000, $a$ 0.05, $\gamma$ 0.2, $c_{2c}$ 0.3, no drift, $\sigma_r$ 0.02); in parentheses the interpolated crossing. $^{\dagger}$: the base itself fails and the value shown is the improvement needed; $^{*}$: passes over the whole grid; n/r: not reachable on the grid. Last column: the joint specification (every axis at its threshold at once), mean gap over 3 seeds, round 1 or the tightened round 2.

| Size | Levels | Asym. $a$ | Nonlin. $\gamma$ | Noise $c_{2c}$ | Retention $\tau$ | Ref. err. $\sigma_r$ | Joint spec |
|---|---|---|---|---|---|---|---|
| CIFAR 16 | n/r | n/r | n/r | n/r | n/r | $\leq$ 0 (0.0104)$^{\dagger}$ | – |
| CIFAR 32 | $\geq$ 2000 (1488)$^{\dagger}$ | n/r | n/r | $\leq$ 0 (0.17)$^{\dagger}$ | n/r | $\leq$ 0 (0.0179)$^{\dagger}$ | – |
| CIFAR 64 | n/r | n/r | n/r | n/r | n/r | n/r | – |
| LM 64 | $\geq$ 500 (292) | $\leq$ 0.2 (0.231) | $\leq$ 0.2 (0.379) | $\leq$ 1 (1.12) | $\geq$ 1e4 (7e3) | $\leq$ 0.02 (0.0239) | r1: 4.74 pass |
| LM 128 | $\geq$ 1000 (612) | $\leq$ 0.2 (0.21) | $\leq$ 0.2 (0.281) | $\leq$ 0.3 (0.727) | $\geq$ 1e4 (8e3) | $\leq$ 0.02 (0.021) | r1: 2.45 pass |
| LM 256 | n/r | n/r | n/r | n/r | n/r | n/r | – |

Table 22: Version 1 (superseded). Degradation thresholds (descriptive, added after the one-at-a-time data were seen; DEVIATIONS-PAPER.md D1): the weakest grid value of each parameter whose gap stays within the line (2 pp CIFAR; 5% LM) of the protonic base device's own gap (column 3), all other parameters at the base; interpolated crossing in parentheses; $^{*}$ the whole grid stays within; n/r the nearest grid step already leaves the band.

| Alg. | Size | Base gap | Levels | Asym. $a$ | Nonlin. $\gamma$ | Noise $c_{2c}$ | Retention $\tau$ | Ref. err. $\sigma_r$ |
|---|---|---|---|---|---|---|---|---|
| SGD | CIFAR 32 | 19.80 | $\geq$ 1000 (520) | $\leq$ 0.05 (0.0813) | $\leq$ 0.2 (0.266) | $\leq$ 1 (1.3) | $\geq$ 1e4 (7e3) | $\leq$ 0.1 (0.15) |
| SGD | LM 128 | 39.85 | $\geq$ 200 (143) | $\leq$ 0.4$^{*}$ | $\leq$ 0.2 (0.423) | $\leq$ 4$^{*}$ | $\geq$ 1e4 (3e3) | $\leq$ 0.5$^{*}$ |
| TTv2 | CIFAR 16 | 4.03 | $\geq$ 100$^{*}$ | $\leq$ 0.2 (0.204) | $\leq$ 0.5 (0.606) | $\leq$ 0.3 (0.789) | $\geq$ 1e5 (7e4) | $\leq$ 0.02 (0.0228) |
| TTv2 | CIFAR 32 | 2.39 | $\geq$ 500 (265) | $\leq$ 0.2 (0.209) | $\leq$ 0.5 (0.578) | $\leq$ 2 (2.31) | $\geq$ 1e5 (2e4) | $\leq$ 0.02 (0.0241) |
| TTv2 | CIFAR 64 | 7.32 | $\geq$ 100$^{*}$ | $\leq$ 0.2 (0.203) | $\leq$ 2$^{*}$ | $\leq$ 2 (2.62) | $\geq$ 1e5 (4e4) | $\leq$ 0.02 (0.0244) |
| TTv2 | LM 64 | 3.18 | $\geq$ 200 (130) | $\leq$ 0.2 (0.284) | $\leq$ 0.5 (0.717) | $\leq$ 2 (2.03) | $\geq$ 1e4 (5e3) | $\leq$ 0.02 (0.0326) |
| TTv2 | LM 128 | 4.25 | $\geq$ 200 (167) | $\leq$ 0.2 (0.276) | $\leq$ 0.5 (0.705) | $\leq$ 1 (1.67) | $\geq$ 1e4 (6e3) | $\leq$ 0.02 (0.028) |
| TTv2 | LM 256 | 16.29 | $\geq$ 100$^{*}$ | $\leq$ 0.4$^{*}$ | $\leq$ 2$^{*}$ | $\leq$ 2 (2.64) | $\geq$ 1e4 (4e3) | $\leq$ 0.02 (0.0351) |

Table 23: Version 1 (superseded). Pairwise sensitivity (E5): the three largest interactions per task, interaction = gap(a,b) − [gap(a) + gap(b) − gap(base)].

| Task | Pair | Gap(a,b) | Singles | Interaction |
|---|---|---|---|---|
| CIFAR | asym=0.1, tau_steps=1e3 | 46.64 | 1.82 + 33.96 | 13.25 |
| CIFAR | asym=0.2, tau_steps=1e3 | 40.24 | 1.92 + 33.96 | 6.75 |
| CIFAR | gamma=0.5, tau_steps=1e3 | 41.17 | 3.55 + 33.96 | 6.05 |
| LM | c2c=1, tau_steps=1e3 | 76.94 | 5.27 + 33.74 | 42.18 |
| LM | gamma=0.5, tau_steps=1e3 | 76.63 | 6.28 + 33.74 | 40.85 |
| LM | asym=0.1, tau_steps=1e3 | 73.70 | 4.09 + 33.74 | 40.11 |

Table 24: Version 1 (superseded). System energy per weight per row on the character LM (E6), in pJ: forward and backward MVMs at the measured 12.4 TOPS/W of [36], device pulses (simulated count × published or estimated energy per pulse), pulse lines and generation, and the TT-family reads. Ratio = H100 at its measured utilisation over the analog total; the range sets every input at its low or high end (records/ENERGY-SYSTEM.json).

| Device | Algorithm | Pulses/w/row | MVM | Device | Lines | TT | Ratio | Range |
|---|---|---|---|---|---|---|---|---|
| Li-ECRAM (Tang 2018) | AGAD | 0.034 | 0.323 | 0.0000 | 0.0013 | 0.00003 | 32.4 | 16.2–234 |
| Li-ECRAM (Tang 2018) | c-TTv2 | 0.042 | 0.323 | 0.0000 | 0.0013 | 0.00003 | 32.3 | 16.2–234 |
| Li-ECRAM (Tang 2018) | SGD | 0.001 | 0.323 | 0.0000 | 0.0012 | 0.00000 | 32.4 | 16.2–234 |
| Li-ECRAM (Tang 2018) | TTv2 | 0.038 | 0.323 | 0.0000 | 0.0013 | 0.00003 | 32.4 | 16.2–234 |
| MO-ECRAM (Kim 2019) | AGAD | 0.267 | 0.323 | 0.0003 | 0.0015 | 0.00003 | 32.3 | 16.2–232 |
| MO-ECRAM (Kim 2019) | c-TTv2 | 0.339 | 0.323 | 0.0003 | 0.0015 | 0.00003 | 32.3 | 16.2–232 |
| MO-ECRAM (Kim 2019) | SGD | 0.007 | 0.323 | 0.0000 | 0.0013 | 0.00000 | 32.4 | 16.2–234 |
| MO-ECRAM (Kim 2019) | TTv2 | 0.300 | 0.323 | 0.0003 | 0.0015 | 0.00003 | 32.3 | 16.2–232 |
| Protonic (Onen 2022) | AGAD | 0.026 | 0.323 | 0.0001 | 0.0038 | 0.00003 | 32.1 | 15.8–228 |
| Protonic (Onen 2022) | c-TTv2 | 0.035 | 0.323 | 0.0001 | 0.0038 | 0.00003 | 32.1 | 15.8–228 |
| Protonic (Onen 2022) | SGD | 0.001 | 0.323 | 0.0000 | 0.0018 | 0.00000 | 32.3 | 16.1–233 |
| Protonic (Onen 2022) | TTv2 | 0.033 | 0.323 | 0.0001 | 0.0037 | 0.00003 | 32.1 | 15.8–228 |
| Memistor (Widrow 1960, as claimed) | AGAD | 0.017 | 0.323 | 262122.1163 | 0.0012 | 0.00003 | 4e-05 | 2.69e-05–4.26e-05 |
| Memistor (Widrow 1960, as claimed) | c-TTv2 | 0.022 | 0.323 | 323881.6451 | 0.0012 | 0.00003 | 3.24e-05 | 2.18e-05–3.45e-05 |
| Memistor (Widrow 1960, as claimed) | SGD | 0.002 | 0.323 | 30015.2982 | 0.0012 | 0.00000 | 0.000349 | 0.000235–0.000372 |
| Memistor (Widrow 1960, as claimed) | TTv2 | 0.022 | 0.323 | 326250.3670 | 0.0012 | 0.00003 | 3.21e-05 | 2.16e-05–3.42e-05 |
| W/YSZ/$WO_x$ (Chen 2023) | AGAD | 0.007 | 0.323 | 0.0000 | 0.0015 | 0.00003 | 32.3 | 16.2–233 |
| W/YSZ/$WO_x$ (Chen 2023) | c-TTv2 | 0.010 | 0.323 | 0.0000 | 0.0015 | 0.00003 | 32.3 | 16.2–233 |
| W/YSZ/$WO_x$ (Chen 2023) | SGD | 0.000 | 0.323 | 0.0000 | 0.0013 | 0.00000 | 32.4 | 16.2–234 |
| W/YSZ/$WO_x$ (Chen 2023) | TTv2 | 0.009 | 0.323 | 0.0000 | 0.0015 | 0.00003 | 32.3 | 16.2–233 |

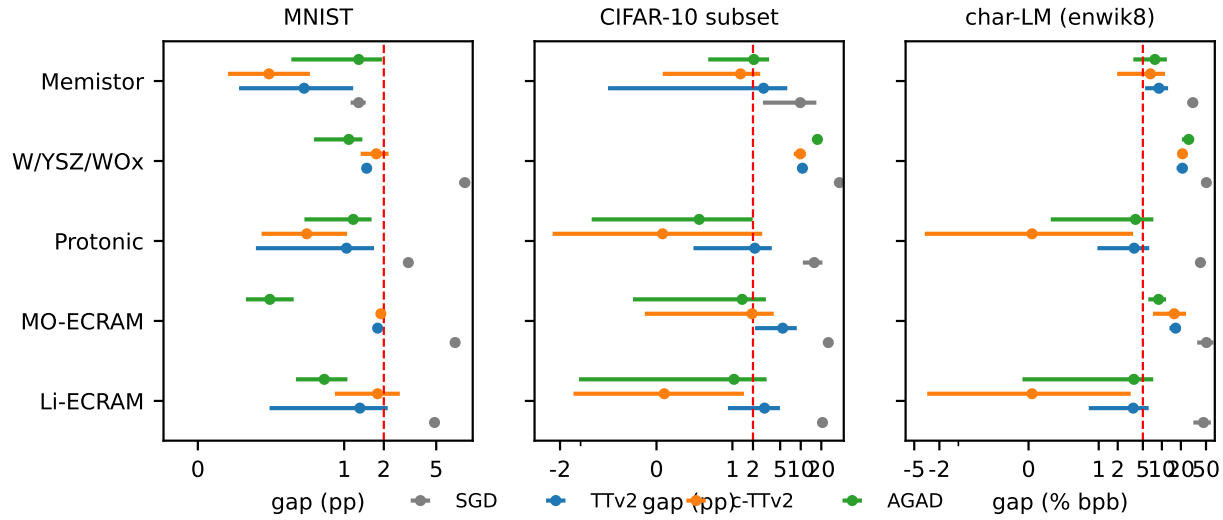



Figure 3: Version 1 (superseded). Original simulator: gap to the exact-update baseline (mean and 95% interval over 5 seeds; symmetric-log axis); the dashed line is the gap line.

# F Revision history and response to the independent review

**History.** Version 1 (October 25, 2026) reported the experiments of Appendix E. An independent technical review (October 26, 2026) recommended major revision with thirteen findings. Version 2 answered them; the full response, finding by finding, with the checks that confirmed or qualified each one, is `RESPONSE-TO-REVIEW.md` in the release. The revision plan was committed before any revision run (`PLAN-REVISION.md`); its deviations R1–R6 are recorded in `DEVIATIONS-PAPER.md`. Version 3 (October 26, 2026) answered a second review (eight findings; `RESPONSE-TO-REVIEW-2.md`, plan `PLAN-REVISION-2.md`): the pulse-fidelity experiments of Section 4.3, the recentred fixed-symmetry control, the width contrasts paired by seed and the revised settings tables. Version 4 (October 27, 2026) answers a third review (five findings; `RESPONSE-TO-REVIEW-3.md`, plan `PLAN-REVISION-3.md`, committed before its runs): the per-sample stream is compared with the literal one paired by draw, with bootstrap bounds, in a desk rerun of the update probes that keeps the draws; the cross-cell dependence is measured by per-pair correlations that cannot cancel; the training comparison reports both arms' labels and every rank reversal; the open training rerun of the one material cell was done; and an interval that includes the width margin is no longer read as growth below it. Version 5 (October 27, 2026; this paper) answers a fourth review (three findings; `RESPONSE-TO-REVIEW-4.md`, plan `PLAN-REVISION-4.md`, committed before its runs). Version 4's five-seed rerun of the material SGD cell had joined, by seed only, two stream runs whose initialisation and example order differed from their controls' (the random draws depend on the GPU model), and it had put a seed-0 difference down to GPU rounding; its figures (stream gap 11.04 pp, change -0.47 pp) are withdrawn as an unmatched, mixed-hardware diagnostic. The cell was rerun as matched triples on one GPU, and the analysis now refuses a seed whose arms do not share both hashes. The saved pulse draws joined the deposit with a command that rebuilds the probe tables from them, the probe table was split in two for legibility, and the abstract states that its probe count is a point-estimate count.

**What was repaired.**

1. **Random streams.** The initialisation, the data order, each tile's device population and the pulse/noise draws now have separate generators; every record carries the hashes of its logical initialisation and data order, and every arm of a seed shares them (checked). The old single stream is kept as a mode that replays the original simulator bit for bit.

2. **The reset.** The original reset moved $A$ to its symmetry point and charged 62 pulses it never applied; applied through the device model those pulses leave 97.55 % of the displacement (from $u = 0.8$ the state only reaches 0.7835, against a symmetry point of 0.125; about 11,513 alternating pulses would be needed for a hundredfold reduction). The repaired pulse reset applies and counts every pulse; the ideal reset is reported only as a labelled variant.

3. **The AIHWKIT presets** are mapped one to one (shifted bounds, bound-scaled slopes, slope spread), and their pulse responses were compared with AIHWKIT's own (Section 4.1): the mapping is accepted.

4. **Measured, not assumed:** the clipping of the pulse probabilities and the lost update are recorded per layer.

The 8 checks of the repairs each fail on the old behaviour or a planted fault (8 faults; ALL_OK on the desk and on every pod).

**How much the repairs moved the original conclusions (diagnostic D1).** Table 25 runs the original and the repaired simulator side by side at the original selections, 3 seeds, on MNIST and a 10-epoch CIFAR slice (234 runs).

- **The random streams** moved individual cells by up to 1.68 pp on MNIST and 3.89 pp on CIFAR (material, i.e. more than 1 pp or a label change, in 4 of 8 and 7 of 8 cells). With three seeds this is within the noise: the repaired FP-SGD alone spans 52.0–57.2 % on the CIFAR slice. The matched controls did not reduce the paired-gap spread here (median 1.78 against 0.76 pp).
- **The reset** moved cells by up to 1.38 pp on MNIST and 4.51 pp on CIFAR. The ideal reset gave no hidden advantage: a physical reset (pulse, alternating pairs or none) had an equal or smaller gap in 19 of 24 cells.
- **The mapping** moved Li-ECRAM by at most 0.52 pp on MNIST and 1.08 pp on CIFAR.
- **Unchanged:** plain SGD fails by far (MNIST 3.09–6.99 pp, CIFAR slice 10.35–17.87 pp).

**Wording withdrawn or narrowed.** "Near-digital" is not used. "Plain SGD fails on every device" became "every plain-SGD arm at the FP-selected learning rate exceeded the margin". "The gap does not grow with width" became the equivalence statement of Section 4.4. "Two device properties decide" and "a few hundred levels suffice" became conditional statements about the sweeps of Section 4.5. "The memistor lacked density, speed and energy, not accuracy" became "an optimistic historical-device scenario attained useful accuracy in this simulator". "32× today with the periphery counted" became the conditional projection of Section 5. "A bench replication" became "a proposed, unbuilt bench replication protocol". "Every matrix is analog" became "the learned weight matrices use the device model; attention products and other operations are digital".

**Version 3's update probes (superseded).** Table 26 is version 3's analysis of the update probes (pod sitting r6). Its sampling floor for the per-sample stream added the two methods' variances as if their draws were independent, although the per-sample stream replays the literal stream's line firings, so the floor was too large and every excess of the per-sample stream came out as zero. Version 4 replaces it by the paired analysis of Section 4.3 (Table 7). The surrogate's verdicts did not depend on this floor.

Table 25: Diagnostic D1: mean gap to FP-SGD of the same simulator (pp, 3 seeds; P/F/U against 2 pp) at the original selections. Columns: orig. = the original simulator; ideal = the repaired random streams with the original ideal reset; then the reset replaced by the pulse reset (pulse), 31 alternating pulse pairs (alt.), or none; publ. = the published-style TTv2. The AIHWKIT-mapped Li-ECRAM row uses the pulse reset for TTv2 and c-TTv2. CIFAR here is a 10-epoch slice.

| Task | Device | Alg. | orig. | ideal | pulse | alt. | none | publ. |
|---|---|---|---|---|---|---|---|---|
| MNIST | Protonic-insp. | SGD | 3.09F | 4.58U | | | | |
| MNIST | Protonic-insp. | TTv2 | 1.04U | 1.60U | 0.23P | 0.66P | 1.02P | 0.99P |
| MNIST | Protonic-insp. | c-TTv2 | 0.81P | 0.62U | 0.44P | 0.97U | 1.41P | |
| MNIST | Protonic-insp. | AGAD | 1.21P | 0.99P | | | | |
| MNIST | Li (old map) | SGD | 4.79F | 6.47U | | | | |
| MNIST | Li (old map) | TTv2 | 1.33U | 1.19U | 0.39P | 0.36P | 0.66P | 1.43P |
| MNIST | Li (old map) | c-TTv2 | 1.51U | 0.98P | 0.80P | 0.68P | 1.00P | |
| MNIST | Li (old map) | AGAD | 0.96P | 1.00P | | | | |
| MNIST | Li (AIHWKIT) | SGD | | 6.99U | | | | |
| MNIST | Li (AIHWKIT) | TTv2 | | | 0.40P | | | |
| MNIST | Li (AIHWKIT) | c-TTv2 | | | 0.70P | | | |
| MNIST | Li (AIHWKIT) | AGAD | | 0.57P | | | | |
| CIFAR-10ep | Protonic-insp. | SGD | 14.24F | 10.35F | | | | |
| CIFAR-10ep | Protonic-insp. | TTv2 | 5.28U | 2.53U | 2.04U | -0.83U | -1.97U | 4.17U |
| CIFAR-10ep | Protonic-insp. | c-TTv2 | 4.59U | 3.41U | 5.97U | -0.43U | 0.65U | |
| CIFAR-10ep | Protonic-insp. | AGAD | -0.70P | -2.94U | | | | |
| CIFAR-10ep | Li (old map) | SGD | 17.69F | 16.91F | | | | |
| CIFAR-10ep | Li (old map) | TTv2 | 4.35F | 2.51U | 2.98U | 0.57U | 0.78U | 5.41U |
| CIFAR-10ep | Li (old map) | c-TTv2 | 6.08U | 2.27U | 1.63U | 1.48U | 0.99U | |
| CIFAR-10ep | Li (old map) | AGAD | 0.39U | -1.34U | | | | |
| CIFAR-10ep | Li (AIHWKIT) | SGD | | 17.87F | | | | |
| CIFAR-10ep | Li (AIHWKIT) | TTv2 | | | 2.37U | | | |
| CIFAR-10ep | Li (AIHWKIT) | c-TTv2 | | | 2.71U | | | |
| CIFAR-10ep | Li (AIHWKIT) | AGAD | | -1.02U | | | | |

Table 26: Version 3 (superseded: its sampling floor for seq treated the paired draws as independent). Update fidelity (experiment A1; Li = Li-ECRAM, MO = MO-ECRAM, Prot. = protonic-inspired): one training batch applied repeatedly from the same state by the aggregate surrogate (sur.), the per-sample stream (seq) and the literal pulse-by-pulse stream (slot). Lost = unsigned requested pulse mass removed by the probability clipping (%); $g$ = the slot stream's realised gain along the requested update; then surrogate / seq against slot: gain difference (%), excess relative mean difference beyond the sampling floor, median per-cell variance ratio; faithful by the pre-registered lines (gain within 10 %, excess $\leq 0.10$, variance ratio in [0.5, 2]; for seq 5 %, 0.05, [0.8, 1.25]).

| Case | Layer | Upd. | Lost | $g$ | $\Delta g$ | excess | var. ratio | faithful |
|---|---|---|---|---|---|---|---|---|
| CIFAR MO AGAD | 3 (128×1024) | 600 | 32.1 | 0.431 | 0.7 / -0.2 | 0.07 / 0.00 | 1.08 / 1.00 | yes / yes |
| CIFAR MO AGAD | 4 (10×128) | 600 | 92.1 | 0.027 | 1.8 / 0.3 | 0.18 / 0.00 | 1.20 / 1.02 | no / yes |
| CIFAR MO AGAD | 3 (128×1024) | 2000 | 15.0 | 0.700 | 1.3 / -0.0 | 0.03 / 0.00 | 1.07 / 1.00 | yes / yes |
| CIFAR MO AGAD | 4 (10×128) | 2000 | 96.5 | 0.004 | 32.7 / 0.6 | 0.37 / 0.00 | 1.22 / 1.00 | no / yes |
| LM Prot. c-TTv2 | 0 (128×256) | 750 | 7.6 | 0.717 | 0.3 / 0.1 | 0.01 / 0.00 | 1.00 / 1.00 | yes / yes |
| LM Prot. c-TTv2 | 9 (256×128) | 750 | 0.0 | 0.972 | -1.4 / -0.5 | 0.00 / 0.00 | 0.71 / 1.01 | yes / yes |
| LM Prot. c-TTv2 | 0 (128×256) | 2250 | 32.5 | 0.547 | -0.1 / 0.0 | 0.04 / 0.00 | 1.02 / 1.00 | yes / yes |
| LM Prot. c-TTv2 | 9 (256×128) | 2250 | 0.0 | 0.882 | 0.2 / 0.3 | 0.00 / 0.00 | 0.73 / 1.01 | yes / yes |
| MNIST Li c-TTv2 | 0 (256×784) | 1000 | 21.6 | 0.759 | 0.1 / -0.0 | 0.03 / 0.00 | 1.05 / 1.00 | yes / yes |
| MNIST Li c-TTv2 | 2 (10×128) | 1000 | 3.7 | 0.990 | -0.0 / -0.0 | 0.00 / 0.00 | 1.26 / 0.98 | yes / yes |
| MNIST Li c-TTv2 | 0 (256×784) | 3500 | 25.7 | 0.696 | -0.3 / -0.0 | 0.01 / 0.00 | 1.09 / 1.00 | yes / yes |
| MNIST Li c-TTv2 | 2 (10×128) | 3500 | 5.0 | 0.922 | -1.1 / 0.1 | 0.00 / 0.00 | 1.55 / 1.00 | yes / yes |